The Impact of Improved Water and Sanitation on Child Nutritional Outcomes in Sub-Saharan Africa: Evidence from DHS and Geospatial Data

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Abstract

This thesis asks whether water and sanitation (WASH) conditions are associated with child anthropometric outcomes (height-for-age, weight-for-age and weight-for-height z-scores) in four Sub-Saharan African countries: Ethiopia, Ghana, Kenya and Nigeria, selected for data availability and recency. The baseline analysis distinguishes a child's own household WASH facilities from the WASH conditions of the surrounding community; a preliminary extension links earlier DHS rounds for three of the four countries to modelled annual gridded local WASH coverage (IHME) for a defined early-life window, addressing the baseline's limitation that WASH is recorded only once, at the survey date. Both analyses are cross-sectional and observational; every estimate is a conditional association, not a causal effect.

The baseline sample pools each country's most recent DHS round (2022–2025). Household WASH, estimated with cluster fixed effects, reaches no significance at the 5% level for any outcome (water–HAZ closest, $p = .062$): inconclusive, not evidence of no association. Community WASH coverage, estimated with region fixed effects, shows a large, significant association before covariate adjustment that attenuates to non-significance once socioeconomic and region controls are added, consistent with, though not proof of, confounding by local development.

The historical extension's primary finding is the pooled established-date water–HAZ association: $\beta = 1.03$ per 10 percentage points (95% CI [0.51, 1.54], $p < .001$), one of six nominally significant cells among 24 with no multiplicity adjustment, sensitive to the pooled weighting convention (0.41, $p = .035$, unweighted), and independently reproduced with a second estimation library. The estimates are weighted conditional associations among eligible measured children, not population-representative, and carry unresolved measurement-weight documentation and other disclosed limitations.

Because the baseline and historical samples differ at once in children, survey rounds, exposure construction and controls, their contrast cannot be attributed to exposure timing; the historical result is preliminary and does not overturn the baseline conclusions. The thesis contributes a side-by-side household/community comparison and a preliminary attempt at genuine early-life exposure timing. Neither analysis identifies a causal policy effect or cost-effectiveness; the associations reported can inform research questions and interpretation, not policy recommendations.

1 Introduction

Child stunting remains one of the most persistent markers of population health disadvantage in Sub-Saharan Africa. Linear growth faltering is concentrated in the first two years of life, is difficult to reverse once established, and its consequences extend well beyond anthropometry to cognitive development, school attainment, and adult earning capacity (Victora et al. 2008; Victora et al. 2010; Black et al. 2013). Water, sanitation, and hygiene (WASH) conditions occupy a central place among the exposures proposed to explain this variation, via several biologically plausible, separately evidenced mechanisms reviewed in full in Section 2.1. Even the strongest recent causal evidence against basic WASH interventions reducing stunting does not dispute this plausibility (Section 2.3); plausibility is not the same claim as an established causal mechanism, and this thesis does not conflate the two. What is considerably less settled is whether the mechanism translates into a *detectable, policy-relevant statistical association* in observational survey data once household wealth and local development are accounted for, and whether a *basic facility-level* WASH classification, rather than a comprehensive intervention package, is the right exposure in which to look for it — the empirical question this thesis addresses.

1.1 Two distinct exposures: household facilities and surroundings

A household's own water source and sanitation facility are not the only WASH-related exposure a child faces: even a child with an improved latrine and piped water can be exposed through shared water sources, open defecation by neighbours, or proximity to inadequately managed waste if the surrounding community lacks similar facilities. *Own-household WASH* and *surrounding-community WASH* correspond to two different exposure pathways, not a modelling convenience. Improved household sanitation is a textbook source of a local environmental externality, since it reduces the pathogen load the household releases into the shared environment, and recent research (Donohue et al. 2023; Skoufias and Vinha 2026) suggests household and community WASH need not move together or carry the same association with child growth. The thesis therefore constructs and compares both: a child's own household classification, and a leave-one-out measure of improved-facility coverage among all *other* households in the same DHS cluster.

1.2 Exposure timing in cross-sectional survey designs

A measurement limitation affects the cross-sectional, repeated-survey-round literature this baseline analysis belongs to (Rakotomanana et al. 2020; Addae et al. 2024): household and community WASH status is recorded once, at the survey date, and applied to every child regardless of age, so a four-year-old's WASH measurement describes conditions years after their nutritionally critical first two years, while an infant's describes conditions close to their own early life. This does not apply to the major WASH trials, which assign exposure near birth and follow growth prospectively (Section 2.3), nor to panel designs tracking indicators over successive rounds (Section 2.2). Age controls and age-interaction tests, also estimated here, describe how the *current* association varies with a child's age today; they do not reconstruct a child's actual early-life environment. The baseline analysis documents this limitation and does not claim to resolve it.

A separate, bounded extension responds to one dimension of it. It does not ask whether assigning a *historical* instead of a *current* exposure changes an otherwise identical estimate — the baseline and historical samples differ too much in other respects to isolate that comparison (Section 6.2). Instead it asks, within its own sample: *is modelled local WASH coverage during a child's own early-life window associated with later anthropometric outcomes?* It matches each child's birth-cohort location to a modelled WASH coverage surface for a defined early-life window, using IHME's gridded 2000–2017 estimates (Local Burden of Disease WaSH Collaborators 2020), giving genuine within-cluster, cross-birth-cohort variation a single survey round cannot offer — an improvement on timing specifically, not a record of a child's actual household facilities. It carries its own separate limitations (Section 1.6; Section 6).

1.3 Research questions and scope

This thesis asks three questions, each posed and answered as an *association under a cross-sectional design*, not a causal effect: (1) is a child's own household water/sanitation status associated with HAZ/WAZ/WHZ after household- and cluster-level confounding is accounted for; (2) is the surrounding community's WASH status, independent of the household, separately associated with these outcomes; (3) within the three historical-sample countries, is modelled local WASH coverage during a child's early-life window associated with these outcomes? The third question is answered only within the historical sample; its answer is not compared with the first two as if that isolated an exposure-timing effect (Section 6.2).

The baseline pools each country's most recent DHS round: Ethiopia (2024–25), Ghana

(2022), Kenya (2022), Nigeria (2024), selected for data availability and recency, not as a representative sample of Sub-Saharan Africa. The historical extension uses an earlier round for three of the four (Ghana 2014, Kenya 2014, Nigeria 2018), linked to IHME's gridded estimates; Ethiopia's 2016 round enters only as a provisional sensitivity analysis, because its interview-date conversion could not be fully validated (Section 3.5, Appendix C). The baseline and historical samples are *not the same children*, use different exposure definitions, and are analysed and reported separately; no result in one adjusts, selects, or validates a result in the other.

1.4 Contribution

This thesis makes two separate contributions in two separate samples, not one combined design. In the *baseline sample*, the contribution is the explicit, side-by-side construction and comparison of own-household and leave-one-out community WASH exposure within one four-country sample and one consistent control set, each exposure estimated under the fixed-effects structure its own construction requires (Sections 4.2–4.3). In the *historical sample*, the contribution is a transparently documented, preliminary attempt to introduce genuine temporal variation in early-life exposure through geospatial matching (Sections 3.5–3.8); it does not reproduce the household-versus-community comparison, since the historical data source does not distinguish a child's household from its surroundings. A particular four-country combination is not, by itself, evidence of substantive novelty, and this thesis does not claim it as such; the methodological limitations of each sample are treated as integral to its interpretation, not as closing caveats.

1.5 A restrained summary of findings

Household-level WASH shows no statistically significant association with any outcome under cluster fixed effects. Community-level WASH coverage shows a large, significant association before covariate adjustment that attenuates to insignificance once socioeconomic and region controls are added, consistent with, though not proof of, confounding by local conditions. Within the historical sample, the pooled established-date water–HAZ association is positive and significant, but sensitive to the pooled weighting convention and carries unresolved documentation issues. None of these results identifies a causal policy effect or cost-effectiveness; Sections 5–6 develop them with full uncertainty, and the Abstract gives exact coefficients.

1.6 Principal limitations, stated at the outset

This is a cross-sectional, associational design throughout, not a panel, difference-in-differences, or instrumental-variables design. The principal limitations, stated in full in Section 6, are unresolved sampling-weight documentation for the historical extension's measured-anthropometry subsample, DHS/IHME input overlap in two of its three countries, current residence as a proxy for historical residence, confidentiality-driven displacement of DHS cluster coordinates, and the small number of countries in both samples.

2 Literature review and conceptual framework

2.1 From biological plausibility to demonstrated association

The hypothesised pathway from WASH to child growth runs through several distinct mechanisms, not one undisputed mechanism: inadequate water and sanitation can increase exposure to faecal–oral pathogens, raising diarrhoeal disease, soil-transmitted helminth infection, and sub-clinical environmental enteric dysfunction (EED), each potentially redirecting energy from linear growth, though EED specifically is the least settled (Cumming and Cairncross 2016). Even the strongest recent evidence against basic WASH interventions reducing stunting does not challenge this plausibility: the authors of the three major trials discussed below state explicitly that "the biological plausibility of WASH as public health interventions is not challenged by these findings" (Cumming et al. 2019), arguing the *specific, basic* interventions tested were insufficient to translate the mechanism into a detectable effect. Black et al. (2013) place WASH-related infection among the proximate determinants of undernutrition in their *Lancet* framework, and Victora et al. (2010) document that growth faltering concentrates from birth to roughly 24 months. This review addresses a narrower question: whether a *facility-type classification* — an "improved" water source or sanitation facility in the WHO/UNICEF JMP sense used throughout — shows a detectable statistical association with child growth once obvious confounders are accounted for, distinct from whether WASH-related infection can plausibly affect growth at all.

2.2 Observational evidence and the confounding problem

A substantial observational literature finds WASH variables associated with child anthropometric outcomes in DHS and related data. Rakotomanana et al. (2020) link JMP water and sanitation indicators to linear growth in children 6–23 months across East African DHS surveys, in the expected direction. Addae et al. (2024), using Ghana's 2017/18 MICS, find household sanitation and socioeconomic status jointly associated with stunting and wasting, and unprotected water associated with wasting but not stunting. Günther and Fink (2010) pool 172 DHS surveys and find improved water and sanitation associated with lower child mortality; Spears (2013) finds cross-country height variation correlates with open-defecation rates.

The central difficulty this thesis faces directly is that WASH status is not randomly assigned: it correlates with wealth, education, and local infrastructure, which independently predict nutrition through channels unrelated to pathogen exposure. Headey and Palloni (2019) address this with a panel of 442 subnational regions across 59 countries, using within-region difference-in-differences. Their results are mixed: water access is insignificant for most outcomes (piped water alone predicts reduced stunting), and sanitation predicts large reductions in diarrhoea and mortality but **not** stunting or wasting once within-region design is used. This null sanitation–stunting finding, once region-level time-invariant confounding is netted out, parallels this thesis's own null household- and community-sanitation findings (Section 5); it is not evidence that WASH associations generally weaken under better controls, which would overstate what one study establishes.

2.3 Credible causal evidence: what the major trials found

Randomised evidence comes primarily from three cluster-randomised trials, described individually because their exposure-timing structure differs materially from the cross-sectional literature above: WASH-Benefits Bangladesh, WASH-Benefits Kenya, and the SHINE trial in Zimbabwe enrolled pregnant women and followed children prospectively for 18–24 months from near birth — unlike the single-survey-date measurement this thesis's own baseline design uses (Section 1.2). The three did not share one factorial design: Bangladesh and Kenya tested water, sanitation, hygiene, combined-WASH and nutrition as **separate arms**; SHINE used a simpler 2×2 design with a single combined WASH package. On linear growth, **every individual WASH-related arm in Bangladesh and Kenya, and SHINE's combined WASH arm, showed no statistically detectable effect** (Cumming et al. 2019). The nutrition arm (no WASH) showed a separate, modest positive effect; combined WASH-plus-nutrition showed

no additional benefit over nutrition alone in any trial. On diarrhoea, results were mixed: a substantial reduction in Bangladesh, none detectable in Kenya or Zimbabwe.

"No statistically detectable effect" is not "proven to be exactly zero": these are adequately powered trials, but a null result reflects the trial's power, and the authors do not generalise the finding to every possible WASH intervention. The consensus statement attributes the null to the limited scope of the interventions tested — household-level "basic" services, not "safely managed" networked services — calling for research on more comprehensive WASH rather than concluding WASH cannot matter (Cumming et al. 2019). This bears directly on this thesis: its facility-status exposure is likewise a classification, not a comprehensive intervention package, so a null or weak association here is consistent with, though it does not confirm, the trial finding, and equally consistent with confounding masking a real effect or this thesis's own limitations (Section 6).

A related quasi-experimental literature targets infant mortality. Geruso and Spears (2015) show Muslim infants in India survive to their first birthday at a higher rate than Hindu infants despite lower wealth, isolating a sanitation-specific channel from the fact that Hindus practise open defecation at a rate roughly 25 points higher nationally — a religious difference unexplained by wealth, and a source of sanitation variation distinct from general local development that this thesis's own four-country comparison cannot replicate. The study is cited for this identification strategy, not as direct evidence about growth, which it does not examine.

2.4 Household versus community exposure

Most DHS-based studies measure WASH at the household level only; a smaller strand examines community-level exposure. Donohue et al. (2023) construct geospatially modelled community sanitation coverage and relate it to stunting in Nigeria and Zambia: a mixed result — significant in Nigeria (adjusted OR = 0.97, 95% CI [0.95, 0.99] per 10-point increase) after household sanitation lost significance, not significant in Zambia for either. Skoufias and Vinha (2026), across twenty Sub-Saharan countries, find no significant water externality but a significant community-sanitation association with fever/diarrhoea and height-for-age — contrasting with this thesis's own fully adjusted community HAZ models (Section 5.2), from a different sample and model. This thesis's leave-one-out construction follows the same logic; no study was found pairing it with a household-level comparison in the same four-country sample. Balk et al. (2004) provide an earlier precedent for DHS-cluster geography as a unit of analysis distinct from the household, for mortality rather than anthropometric outcomes.

A specific identification issue follows from this construction: the leave-one-out rate is a

cluster-level property, so estimating it with cluster fixed effects would remove nearly all its substantive variation, identifying the coefficient almost entirely from the mechanical residual that arises because a different household is excluded from each child's own calculation (Section 4 derives this precisely). This thesis therefore uses region, not cluster, fixed effects for the community specification.

2.5 Exposure timing and geospatial matching precedent

The single-survey-date limitation (Section 1.2) applies to this thesis's baseline design and to Rakotomanana et al. and Addae et al. above, but not to Günther and Fink (mortality outcomes) or Spears (a cross-country correlation), nor to the trials (Section 2.3) or Headey and Palloni's panel design. This is why the historical extension attempts a different, early-life-window exposure assignment within its own sample — not a reason to expect comparing the two samples to isolate an exposure-timing effect (Section 6.2).

Assigning a gridded exposure surface to a survey respondent's location requires handling two documented DHS features: coordinates are randomly displaced for confidentiality (Burgert et al. 2013; Perez-Heydrich et al. 2013), so any extraction method reads pixels around the *recorded*, not true, location; and a containing-pixel extraction (the single pixel a point falls in) differs from a buffer-based average. This thesis uses an area-weighted buffer average primarily and containing-pixel extraction as a labelled sensitivity (Section 3.6; Section 5.8). Johnson, Jacob, and Brown (2013) provide an earlier precedent linking satellite-derived surfaces to DHS clusters for a different outcome. Blom, Ortiz-Bobea, and Hoddinott (2022) match geocoded heat-exposure data to each child's developmental window across five West African countries — the design structure this thesis's historical extension borrows, though their outcome is heat exposure's effect on nutrition, not WASH, and the study is cited only for its matching-method structure.

2.6 This thesis's contribution and the gap it addresses

This thesis combines two elements not found together elsewhere in this review: household and leave-one-out community WASH compared directly within one sample and controls (Section 2.4); and a separate, bounded, preliminary attempt to assign modelled local WASH coverage to each child's early-life window via geospatial matching (Section 2.5), within the three historical-sample countries. The four-country combination alone is not evidence of substantive novelty; the claimed contribution is narrower — a direct, consistently controlled comparison

of the two exposure levels, and a transparent early-life exposure-matching attempt, including its own unresolved issues, that neither Donohue et al. nor Skoufias and Vinha implement, since both examine only current-round community exposure. The gap that remains open, and that this thesis does not claim to close, is the one Headey and Palloni and the trial literature already identify at a larger scale: credibly separating a WASH-specific effect from the broader package of local development that improved facilities tend to accompany. The historical extension narrows only the exposure-timing dimension of that problem; it resolves neither the confounding dimension nor exposure-timing for the baseline sample, since the two samples differ in several other respects at once (Section 6.2).

3 Data and variable construction

This section documents the baseline and historical samples separately: they come from different DHS rounds, use different exposure constructions, and are validated independently, so no claim about one sample applies to the other. Full variable definitions and sample-flow tables are in Appendix A; estimation results are reported in Section 5.

3.1 Baseline sample: unit of analysis, rounds, and pooling

The unit of analysis is the child. The baseline pools each country's most recent DHS round as of data collection: Ethiopia (2024–25), Ghana (2022), Kenya (2022), Nigeria (2024). The pooled, harmonised child file contains 70,231 children across 4,481 DHS clusters and 114 administrative-region fixed-effect cells (one region set per country). This pooled file is **not** the regression sample for any model; it is the universe before outcome-validity and exposure-classifiability restrictions are applied.

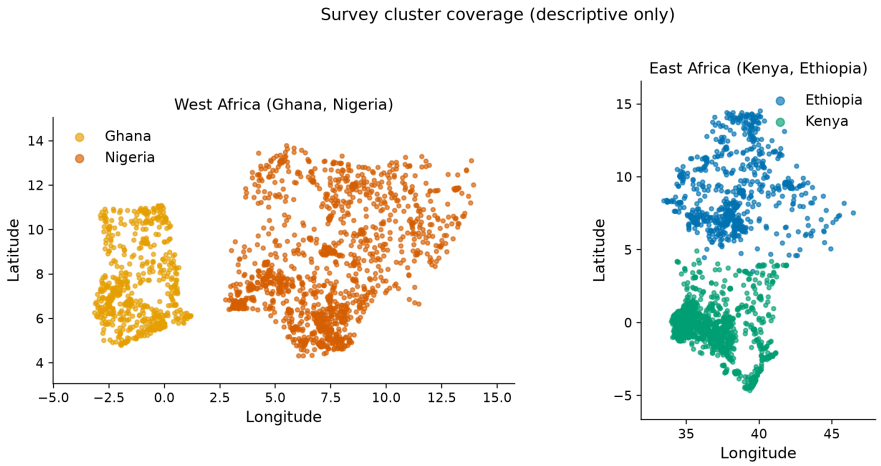


Figure 3.1 — Survey cluster locations, by country (descriptive only; coordinates randomly displaced for confidentiality, Section 3.4).

3.2 Baseline anthropometric outcomes and under-five eligibility

Three outcomes follow standard DHS/WHO practice: height-for-age (HAZ), weight-for-age (WAZ), weight-for-height (WHZ), each a z-score against the WHO Child Growth Standards (WHO Multicentre Growth Reference Study Group 2006). A child contributes to an outcome only if its z-score is flagged valid in the DHS recode (excluding DHS's own special codes 9996–9999 and system-missing values). Valid measurements give smaller outcome-specific samples: 36,985 (HAZ), 37,260 (WAZ), 37,088 (WHZ). Requiring both household and community exposures to be classifiable for the same child gives common samples of 36,040 (HAZ), 36,308 (WAZ), 36,145 (WHZ), across 4,426 clusters. Age is taken uniformly from DHS's standard `b19` field for all four countries; the historical extension (Sections 3.5–3.8) builds its own, differently constructed `age_m` field and the two must not be conflated.

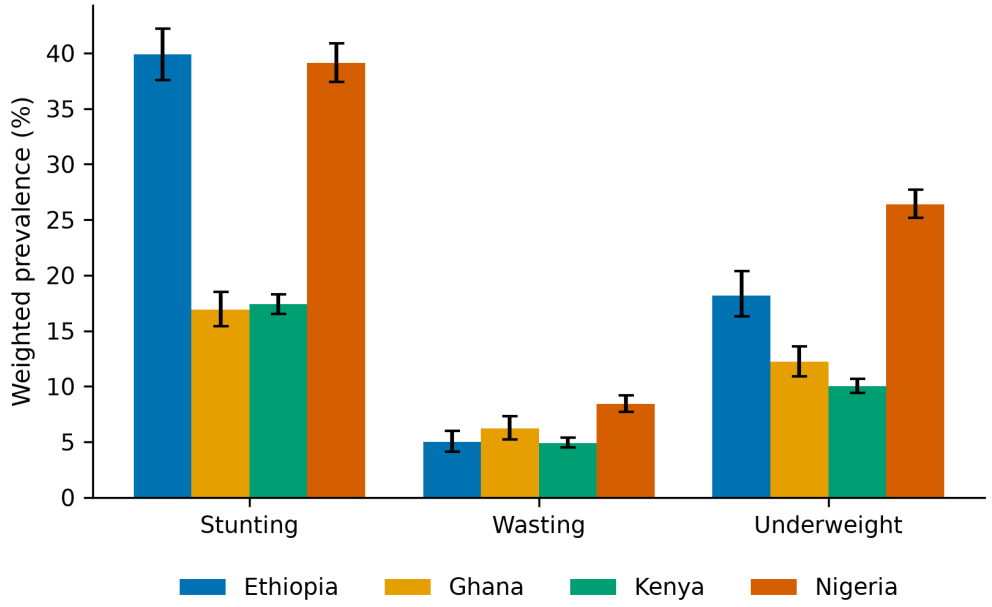


Figure 3.2 — Weighted anthropometric prevalence by country, with 95% confidence intervals (each outcome's own outcome-valid sample, Section 3.2).

Figure 3.2 reports weighted stunting, wasting and underweight prevalence by country, each estimated on that outcome's own outcome-valid sample above, under a stratum/PSU/weight survey design (`strata`, `psu`, the `v005`-derived weight of Section 3.4) with 95% confidence intervals from a logit-transformed interval. This descriptive design is distinct from the PSU-clustered inference the regression models in Section 4 implement; the two are not interchangeable. Stunting is highest in Ethiopia (39.9%) and Nigeria (39.1%), roughly double Ghana (16.9%) and Kenya (17.4%); wasting is lower and more similar across countries (4.9–8.4%); underweight is highest in Nigeria (26.4%) and lowest in Kenya (10.0%). These are descriptive estimates, not regression results; Section 5 models the within-country variation they motivate.

3.3 Baseline WASH exposures: household and leave-one-out community

The baseline sample has four exposures: own-household improved/non-improved water and sanitation, and their leave-one-out community counterparts, following the WHO/UNICEF JMP "improved" classification (the same classification independently verified for the historical extension's gridded product, Section 3.5, Appendix A). Household WASH is a household-level binary indicator. **Community WASH is an unweighted leave-one-out proportion among other sampled, classifiable HR households in the same DHS cluster, excluding the child's own household** — not a census of the community, and not an average over other children.

Households recorded as Other / unclassified (DHS code 96, "other," not assignable to improved or non-improved) or structurally not applicable, and (Ghana's core sanitation variant) an unclassified biodigester, are excluded from both numerator and denominator. The resulting rate is a [0, 1] proportion, missing if no other classifiable household exists in the cluster; coefficients on it are therefore per full unit of coverage (0 to 1), not per 10 percentage points as in the historical tables of Section 5.6. Improved shared sanitation counts as improved via a Basic+Limited collapse; a robustness variant treats Ghana's biodigester differently (Appendix A, Table A.1). A child's own classification is never part of its own community exposure, which makes community WASH vary mainly *between* clusters — Section 4.3 derives why this rules out a cluster-fixed-effects specification for this exposure, unlike for household WASH.

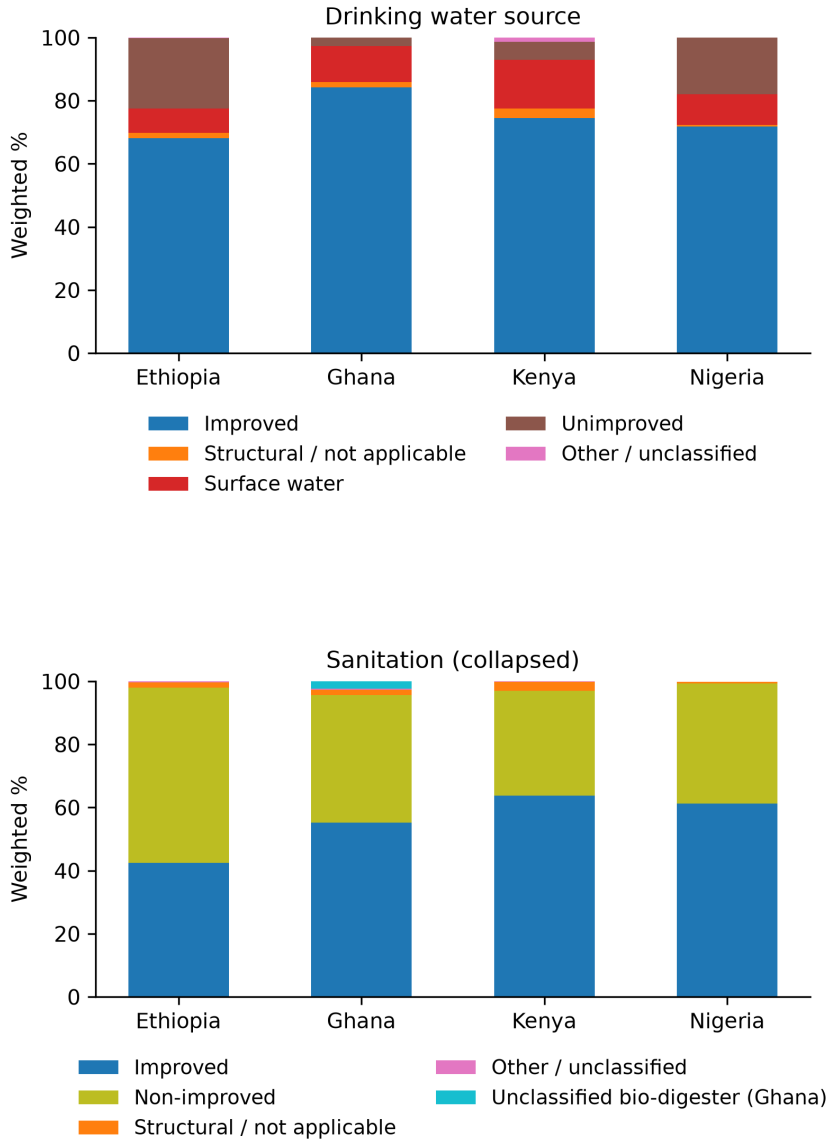


Figure 3.3 — Weighted household WASH characteristics by country, as percentages of each country's full weighted sample (Other / unclassified and structurally-not-applicable cases included in the denominator, not excluded). The sanitation panel uses the collapsed Improved/Non-improved scale: open defecation and other unimproved facility types are folded into Non-improved, not shown as a separate segment.

Figure 3.3 situates these exposures in their country context. Its denominator is each country's full weighted descriptive sample, not the classifiable-household denominator used to construct the regression's leave-one-out community exposure (Section 3.3) — a different quantity for a different purpose. The left panel shows drinking water at full category detail, with surface water and unimproved as their own segments; the right panel shows sanitation on the collapsed Improved/Non-improved scale used in the regressions, where open defecation and other unimproved types are folded into Non-improved rather than displayed separately. Other / unclas-

sified (source responses coded "other" that cannot be assigned to improved or non-improved WASH categories), structurally-not-applicable and (sanitation only) unclassified-biodigester cases remain their own segments in both panels, included in this descriptive distribution but excluded from the classifiable-household denominator used to construct community WASH exposure (Section 3.3). Improved water ranges from 68.1% (Ethiopia) to 84.2% (Ghana). Improved sanitation is lower and ordered differently: 42.5% (Ethiopia), 55.2% (Ghana), 61.2% (Nigeria), 63.8% (Kenya). Only Ethiopia combines high stunting (Figure 3.2) with low sanitation coverage; Nigeria has the second-highest stunting prevalence but sanitation coverage close to Kenya's, so the two figures together do not support a simple cross-country ordering linking WASH coverage to stunting. Sections 4 and 5 examine the within-country, household- and community-level association directly, which this descriptive comparison cannot.

3.4 Baseline covariates, geography, and weights

Covariates include child sex, age in months (`b19`), maternal age, maternal education, literacy, household size, birth order, wealth quintile, and urban/rural residence. The baseline is built from country-specific DHS recode files (the Children's Recode, KR), not IPUMS-DHS, using DHS's own field names throughout — distinct from the historical extension's IPUMS-harmonised variables (Sections 3.5–3.8). The KR file's weight field is `v005`, the woman's individual sample weight (rescaled by the documented 1,000,000 factor), not the household weight `hv005`; this is not `PERWEIGHT`, the different field the historical extension uses. The cluster identifier `v021`, combined with a country prefix into `country_psu`, is analogous in purpose to the historical extension's `IDHSPSU` but constructed independently. DHS cluster coordinates are randomly displaced for confidentiality (urban up to 2 km; rural 99% up to 5 km, 1% up to 10 km) (Burgert et al. 2013) — inconsequential for the baseline models, which use no coordinates directly, but consequential for the historical extension's spatial matching (Section 3.5). Baseline models use the `v005`-derived weight, normalised so each country contributes equal total weight mass regardless of raw sample size (Section 4.5 gives the formula).

3.5 Historical extension: rounds, exposure source, and scope

The historical extension links an earlier DHS round for three countries — Ghana (2014), Kenya (2014), Nigeria (2018) — to IHME's annual gridded estimates of local access to improved water (W_IMP) and sanitation (S_IMP), available 2000–2017 (Local Burden of Disease WaSH Collaborators 2020), verified directly from the primary source (Appendix A, Table

A.2). The extraction computes an **area average of the raster surface within each child's exposure buffer**, not the source study's population-weighted aggregate — a distinction restated wherever a historical coefficient is reported. Ethiopia's 2016 round enters only as a separately labelled, provisional sensitivity analysis (Appendix C), because its Ethiopian-calendar-to-Gregorian interview-date conversion could not be validated to the same standard as the three established-date rounds. The interview date does not place a child's exposure window — windows are placed from the birth date alone (Section 3.7; Appendix C) — but governs eligibility: date validity, birth preceding interview, and (for the two postnatal windows) whether the window had completed by interview. This is a measurement improvement on timing specifically, not a record of a child's actual household facilities, which no available data source can provide.

3.6 Historical spatial extraction: buffers, displacement, and coverage

Each child's exposure buffer is centred on its cluster's displaced coordinates (azimuthal equidistant projection). The primary buffer radius is 2 km (urban) or 5 km (rural). Displacement describes locational *uncertainty*; the buffer radius defines the *area averaged*. A labelled sensitivity repeats the rural assignment at 10 km, and a separate sensitivity uses containing-pixel assignment instead (Section 5.8, Appendix C.4). The raster value is averaged by closed-form disk–polygon overlap, not centroid or nearest-neighbour approximation. A child-year is assigned a valid extraction only when its valid-pixel area meets a documented tolerance (within 10^{-3} of full coverage); partial- or missing-coverage child-years are excluded rather than assigned from incomplete coverage.

3.7 Historical exposure windows and completion rules

Four candidate windows apply to the same annual raster series. **The primary window is the first 12 completed postnatal months** (birth month through birth month + 11, with completion requiring the interview month $t \geq$ birth month + 12; Section 6.4 states why this window was proposed). **The first 24 postnatal months** (birth month through birth month + 23, completion $t \geq$ birth month + 24) is a sensitivity window, not the primary one. Two further alternatives carry no postnatal completion rule: the prenatal window (the nine months before birth, ending at birth) and the birth-year window (the full calendar year of birth). A window spanning two calendar years is weighted by each year's share of months, and every touched year must have a verified raster value. The birth-year window's lack of a completion check is a disclosed limit on that window's eligibility rule, not an oversight. Because IHME's product is annual, the

value used for a window's final touched year can reflect conditions after the interview even when the postnatal completion check is satisfied; this project flags every such case but does not exclude or adjust for it.

3.8 Historical covariates, weights, and the established-date measurement universe

Historical models control for cluster fixed effects, survey-qualified country-by-birth-year terms, child sex, maternal age, maternal education, marital status, and completed age in months. Birth-year terms are each country's own birth-year effects relative to its own earliest observed year, not one pooled reference year; Section 4.6 shows this choice does not itself change the fitted exposure coefficient. The age control uses its own `age_m` field (reported months where the source delivers them, Ethiopia/Nigeria; a calendar-month-difference construction otherwise, Ghana/Kenya), distinct from the baseline's `b19`. Wealth quintile is **excluded from the main specification**, because current wealth may be a post-exposure mediator of historical WASH conditions while also being correlated with the exposure itself — a genuine trade-off, not a one-sided argument: current wealth could instead proxy *prior* socioeconomic circumstances, in which case excluding it leaves that confounding unaddressed. Wealth is included only as a labelled sensitivity (Section 5.8), where the pooled water–HAZ coefficient is essentially unchanged (1.01 vs. 1.03). This differs from, and is more defensible than, claiming cluster fixed effects absorb wealth mechanically: they absorb only what is constant *within* a cluster, confirmed directly since the baseline cluster-fixed-effects model includes and estimates wealth as a standard covariate (Section 4.2).

Weighting uses PERWEIGHT, normalised so each of the three established-date countries contributes an equal total weight within each model's own sample, reported alongside its unweighted counterpart for the headline cell (Section 5.6). No DHS or IPUMS source documents a sampling weight specific to the measured-anthropometry subsample (Appendix A); the working inference, stated wherever historical coefficients are reported, is that within-survey PERWEIGHT normalisation removes a constant measurement-subsample selection factor under random household-level selection — it cannot correct for differential non-response among eligible but unmeasured children. Historical results are therefore reported as **weighted conditional associations among eligible measured children**, not population-representative estimates.

3.9 A measurement-universe discrepancy: Nigeria 2018

Three distinct child populations arise for Nigeria 2018. The official 2018 NDHS report counts 12,806 (unweighted) children under five eligible and measured (men's-questionnaire subsample households). The IPUMS-DHS birth-history extract used here contains 33,924 birth-history records, of which 11,704 are non-missing for anthropometry. The final historical estimation sample (water, HAZ, post-12-month window) retains 6,912 after window-completion, spatial-coverage and singleton-cluster exclusions. These differ by construction — official eligible count, interviewed-women's birth-history children, and the further-restricted estimation sample are three different populations — and the gap between the official count and the extract (12,806 vs. 11,704) is disclosed as an open, unresolved question, not treated as resolved by any estimation choice here.

4 Empirical strategy

Throughout, i indexes a child, h the household, c the DHS cluster, r the administrative region (each cluster belongs to exactly one region), and s a survey round (country). All specifications are estimated and reported as **associational**, not causal, and none is a panel, difference-in-differences, or instrumental-variables design.

4.1 Baseline: household WASH, region fixed effects

$$Y_{ihcr} = \alpha + \beta_W \,\text{Water}_{ih} + \beta_S \,\text{Sanitation}_{ih} + X_i'\gamma + \delta_r + \varepsilon_{ihcr}$$

Water and sanitation are **entered together** in every household specification, never one at a time. X_i is the covariate set of Section 3.4; δ_r is a fixed effect for each of 114 pooled administrative regions, so β_W/β_S are identified from **within-region** variation only. A region contains many DHS clusters, so δ_r still allows clusters to differ in local infrastructure or health-service access, plausibly correlated with both WASH and nutrition and not absorbed here — motivating the cluster-fixed-effects specification below.

4.2 Baseline: household WASH, cluster fixed effects (main specification)

$$Y_{ihc} = \alpha_c + \beta_W \,\text{Water}_{ih} + \beta_S \,\text{Sanitation}_{ih} + X_i'\gamma + \varepsilon_{ihc}$$

α_c, a fixed effect for each cluster, identifies β_W/β_S from differences in household WASH *among households within the same cluster*. It absorbs what every household in a cluster shares, but not differences between households within it — wealth, education, or unobserved behaviour can still differ and keep this specification associational. Urban/rural residence is mechanically absorbed by α_c, since every cluster is entirely one or the other. **Wealth quintile is estimated as part of the covariate set, not omitted**: fixed effects absorb only what is constant within a cluster, and wealth varies within clusters just as WASH status does (31.6 percent of clusters vary in own-water status, 50.6 percent in own-sanitation, the variation identifying the household coefficients; a comparable check confirms wealth varies too), so there is no mechanism to remove it.

4.3 Baseline: community WASH, region fixed effects

$$Y_{ihcr} = \alpha + \beta_W \bar{C}^{water}_{-h,c} + \beta_S \bar{C}^{san}_{-h,c} + X_i'\gamma + \delta_r + \varepsilon_{ihcr}$$

The leave-one-out community rates (Section 3.3) are again entered together, including at Table 5.2's "minimally adjusted" stage. These are cluster-level, not child-level, properties, so β_W/β_S are identified mainly by comparing *clusters* within the same region — an associational estimate of a contextual exposure, not a causal spillover effect.

Why cluster fixed effects are not used here. The leave-one-out rate's within-cluster variation comes only from which single household is excluded from each child's own calculation — a mechanical artefact. Its measured within-cluster standard deviation is approximately 0.007 (water) and 0.011 (sanitation), against an overall standard deviation of approximately 0.35 for both, so a cluster fixed effect would absorb nearly all genuine between-cluster variation. The historical extension (Section 4.6) differs: a real time dimension (birth-cohort variation within cluster) gives its cluster fixed effects genuine, non-mechanical variation to work with — a design consequence of the leave-one-out construction, not an inconsistency.

4.4 Baseline: household and community WASH, jointly; current-age heterogeneity

$$Y_{ihcr} = \alpha + \beta_{HW}\,\text{Water}_{ih} + \beta_{HS}\,\text{Sanitation}_{ih} + \beta_{CW}\,\bar{C}^{water}_{-h,c} + \beta_{CS}\,\bar{C}^{san}_{-h,c} + X_i'\gamma + \delta_r + \varepsilon_{ihcr}$$

All four WASH terms are entered together; each coefficient is conditional on the other three. Because household and community water are fairly closely correlated, this is reported as a secondary decomposition, not a preferred model.

$$Y_i = \alpha + \beta_W \,\text{Water}_i + \beta_S \,\text{Sanitation}_i + \sum_{a \neq \text{ref}} \theta_a \,\text{AgeBand}_{ia}$$
$$+ \sum_{a \neq \text{ref}} \lambda_a^W \,(\text{Water}_i \times \text{AgeBand}_{ia}) + \sum_{a \neq \text{ref}} \lambda_a^S \,(\text{Sanitation}_i \times \text{AgeBand}_{ia}) + X_i'\gamma + \text{FE} + \varepsilon_i$$

AgeBand_{ia} are six current-age bands (0–5 to 48–59 completed months, 12–23 as reference, fixed before any result was examined), replacing the continuous age control. Three joint Wald tests (χ^2) are reported per outcome and fixed-effect structure: water-specific (5 restrictions), sanitation-specific (5), and combined (10). These test whether the survey-date WASH association differs across children's *current* ages (Section 1.2) — not a reconstruction of an older child's actual infancy exposure, and kept distinct from the historical extension's exposure-timing design.

4.5 Inference, weighting, and the pooled estimand

Weighting. Every country is given equal total weight in a pooled model of size N across M countries: $w_i = \text{raw}_i / \sum_{j \in \text{country}(i)} \text{raw}_j$, so each country's weights sum to 1. raw_i is the sample's own native weight field (`v005`-derived for the baseline, Section 3.4; `PERWEIGHT` for the historical extension, Section 3.8), computed within each model's own final sample after applicable exclusions. The historical pipeline additionally multiplies by a constant that only rescales reporting with no effect on any coefficient, SE, CI, or p-value (Appendix B.1). This is an authorised, disclosed working choice, not a correction for sampling design, and does not define the pooled coefficient as any simple average.

The estimand is the single common slope of one pooled regression, not an average of separately estimated country slopes. Equal weight *totals* do **not** imply equal influence: by Frisch–Waugh–Lovell logic, a country's influence on the pooled slope also depends on its residual exposure variation after controls. This thesis's own historical pooled water–HAZ estimate illustrates it: the unweighted average of the three countries' own coefficients is 0.58, while the pooled coefficient is 1.03 (Section 5.6), consistent with Kenya's residual exposure variation being roughly a third of Nigeria's and under half of Ghana's, so its imprecise, negative estimate receives little weight despite equal weight totals. Historical pooled coefficients are reported alongside their country-specific components (Tables 5.5–5.6); baseline pooled models (Sections 5.1–5.3) are not broken out by country in the same way.

Estimator and inference. Cluster fixed effects (household and historical models) are implemented as weighted within-cluster demeaning, equivalent to weighted least squares with cluster dummies, using the cluster-robust CRV1 sandwich with small-sample correction $c =$

$\frac{G}{G-1} \times \frac{N-1}{N-K}$. The two pipelines differ in reference distribution: the historical pipeline uses Student $t(G-1)$; the baseline pipeline (`statsmodels` (Seabold and Perktold 2010)) uses the normal (z) distribution (`use_t: False`, confirmed to ten decimal places, Appendix B.1), as do the region-fixed-effects models. This thesis's primary baseline cluster-FE p-values (water–HAZ .062, sanitation–HAZ .548) are therefore normal-reference values; a more conservative convention penalising degrees of freedom for absorbed cluster dummies gives wider sensitivity p-values (.080, .574), reported as such wherever cited (Appendix B.1).

Singleton clusters and rank. The historical pipeline explicitly drops clusters contributing only one observation and reports the count with every model; the baseline pipeline's code contains no corresponding step, so no baseline singleton count exists — a disclosed description gap, not an assumption it is inconsequential. Every design matrix is checked for full column rank after the fixed-effects transformation; the historical pipeline drops any fully absorbed column by name, and the baseline pipeline's only pre-specified exclusion (`residence` under cluster fixed effects) is stated in Section 4.2. A well-defined, full-rank coefficient is a computational property, not a causal one.

4.6 Historical extension: cluster fixed effects with cohort controls

$$y_{ic} = \alpha_c + \beta\, E_{ic}(w,p) + \lambda\, \text{age}_{ic} + \gamma' X_{ic} + \delta_{s,b_{ic}} + \varepsilon_{ic}$$

$E_{ic}(w,p)$ is the annual-weighted IHME mean (product p, window w) for child i in cluster c, 0–100 scale (Sections 3.5–3.7); β is reported per 1 and per 10 points. $\delta_{s,b}$ is a survey-specific birth-year term, each country's own birth-year effects relative to its own earliest observed year, not one pooled reference year. An early independent-reconstruction attempt used a single pooled reference year and omitted the age-in-months control, producing a materially different coefficient (1.385 vs. the correct 1.026); a dedicated diagnostic isolates the age-in-months omission as the sole cause, the two birth-year schemes being equivalent nuisance specifications confirmed to floating-point precision (Appendix B.2). X_{ic} includes child sex, maternal age, maternal education, and marital status; wealth is excluded for the reason given in Section 3.8, not because cluster fixed effects absorb it.

Why cluster fixed effects are appropriate here but not for community WASH (Section 4.3). $E_{ic}(w,p)$ varies genuinely across birth cohorts *within* the same cluster — real temporal variation, not a mechanical construction artefact. This variation, quantified as the residual exposure standard deviation after controls (Section 5.6), ranges from 0.20 to 0.88 percentage points by country and product — modest relative to the 10-point reporting scale, which this thesis does not interpret as a typical within-cluster change. Weighting follows the same equal-

country-total formula ($M = 3$), and inference uses the Student $t(G-1)$ distribution the historical pipeline implements; singleton clusters are dropped and counted by its own explicit filtering step.

5 Results

Every table in this section is taken directly from the authoritative, validated output files listed in Appendix A; no coefficient was computed for this chapter. Coefficients are reported with 95% confidence intervals, except Table 5.4 (coefficients and p-values only). P-values are shown to three decimal places when at least .0005, and as "<.001" otherwise; full precision is stored in the authoritative files. Where a model has two inference conventions, both are labelled (Section 4.5), and prose uses the primary convention unless the sensitivity is named.

5.1 Household WASH

Table 5.1 — Household WASH: region-FE and cluster-FE results.

Outcome	Water β [95% CI], *p*	Sanitation β [95% CI], *p*	N	Clusters
Region FE				
HAZ	−0.035 [−0.110, 0.040], .357	0.022 [−0.045, 0.089], .522	36,040	4,426
WAZ	0.021 [−0.038, 0.081], .487	0.045 [−0.010, 0.100], .109	36,308	4,426
WHZ	0.050 [−0.015, 0.116], .130	0.046 [−0.011, 0.103], .116	36,145	4,426
Cluster FE (main household specification; PRIMARY inference convention)				
HAZ	−0.085 [−0.175, 0.004], .062	0.026 [−0.058, 0.109], .548	36,040	4,426
WAZ	−0.029 [−0.100, 0.043], .435	0.052 [−0.015, 0.120], .130	36,308	4,426
WHZ	0.012 [−0.062, 0.085], .758	0.053 [−0.014, 0.119], .122	36,145	4,426

Household water and sanitation are entered together in every model (Sections 4.1, 4.2). No household coefficient reaches significance at the 5% level in the cluster-fixed-effects specification, this thesis's main household result and the one least exposed to cluster-level confounding; water–HAZ crosses the 10% threshold. Non-significance is read as inconclusive, not as proof no association exists. Under the more conservative FULL_DUMMY_DF sensitivity, the cluster-FE p-values widen to .080 (water–HAZ) and .574 (sanitation–HAZ) — the same conclusion, using the normal reference distribution the baseline pipeline computes (Section 4.5), not the Student $t(G-1)$ distribution used for the historical extension.

5.2 Community WASH: the adjustment path

Table 5.2 — Community WASH: the four-stage adjustment path (HAZ).

Stage	Water β [95% CI], *p*	Sanitation β [95% CI], *p*
Minimally adjusted	0.447 [0.311, 0.584], <.001	0.472 [0.356, 0.587], <.001
+ child controls	0.395 [0.263, 0.527], <.001	0.452 [0.340, 0.565], <.001
+ socioeconomic status	0.127 [0.010, 0.244], .034	0.008 [−0.117, 0.133], .900
+ SES + region FE (fully adjusted)	0.033 [−0.083, 0.150], .573	0.008 [−0.113, 0.129], .894

Note. HAZ; N = 36,985, 4,434 clusters, all stages; coefficients per full [0, 1] unit of community coverage (not per 10 points). Slightly larger than Table 5.1/5.3's 36,040-child sample, since this specification does not require household WASH to be classifiable. Table 5.3's joint model, on the common sample, is this thesis's like-for-like household/community comparison.

The community coefficients, entered together at every stage, are large and highly significant before adjustment, attenuating sharply as controls are added; neither is significant fully adjusted. This pattern is consistent with confounding by local socioeconomic conditions, but does not, alone, establish it as the sole explanation — a change in measurement structure as controls absorb variance could produce the same pattern. The same four-stage pattern holds for WAZ and WHZ, except WHZ's water association is already small and insignificant before adjustment (0.033, $p = .545$). Across all three outcomes, the fully adjusted specifications (here and Table 5.1) show no consistent WASH–outcome association — inconclusive evidence, not proof WASH has no true relationship with child growth.

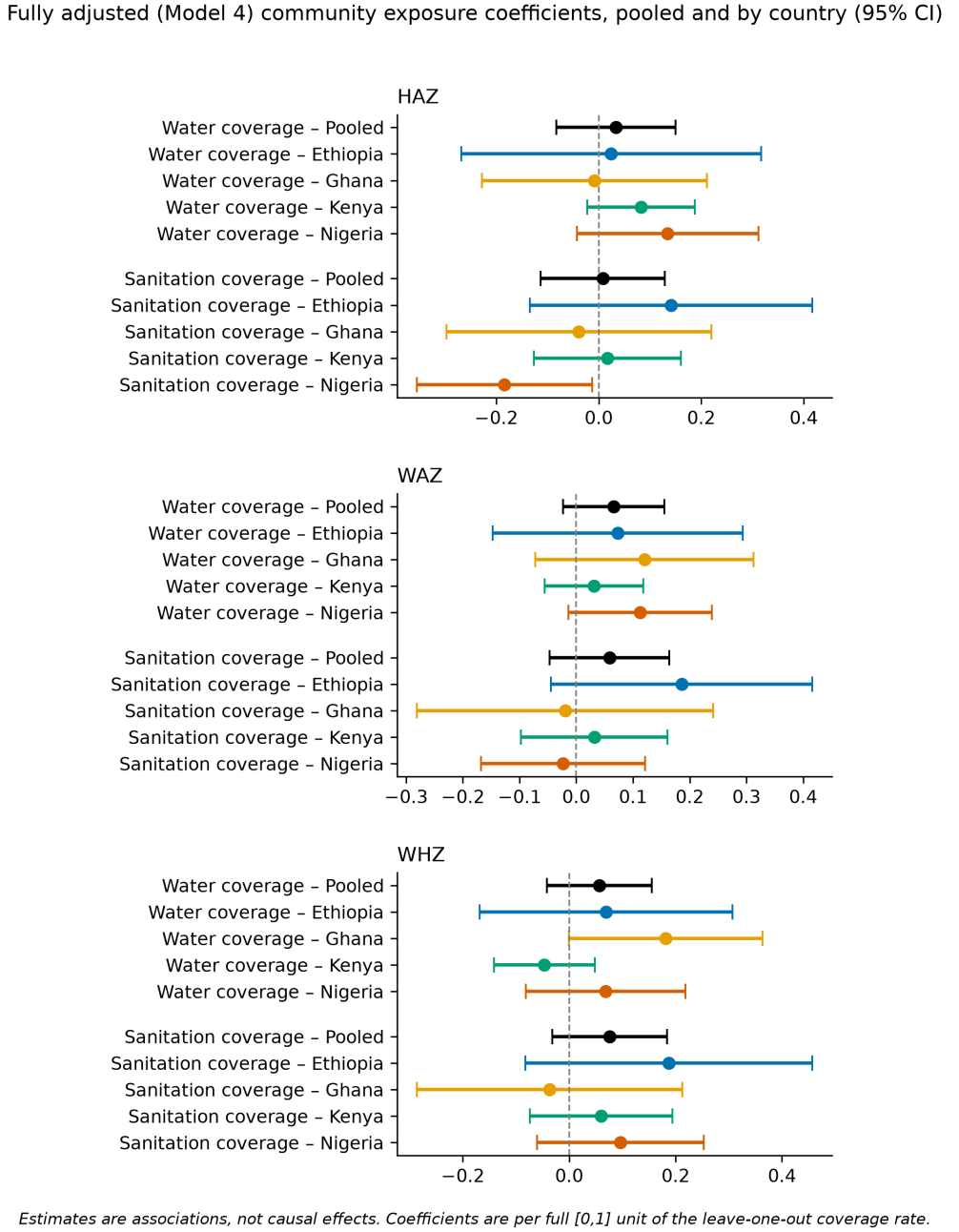


Figure 5.1 — Fully adjusted community water and sanitation coefficients (Table 5.2's final stage), pooled and by country, with 95% confidence intervals.

Figure 5.1 breaks the fully adjusted stage out by country, for the same model, not a separate adjustment path. Of 30 country–outcome–exposure cells, one reaches significance at 5%: Nigeria's community sanitation–HAZ coefficient (−0.184, $p = .035$), already in Table 5.4. Several non-significant cells have comparable or larger point estimates — Ethiopia's sanitation coefficients for WAZ (0.186) and WHZ (0.187) exceed Nigeria's in magnitude — but wider confidence intervals that include zero, consistent with Ethiopia's smaller country subsample; their non-significance reflects imprecision, not a smaller true association, and the pooled null

in Table 5.2 should not be read as averaging a uniformly small set of country effects.

5.3 Joint household/community model

Table 5.3 — Joint household/community model: all four WASH terms entered together.

Exposure	β [95% CI]	*p*
HAZ		
Household water	−0.085 [−0.167, −0.003]	**.043**
Household sanitation	0.024 [−0.047, 0.096]	.507
Community water	0.120 [−0.008, 0.248]	.066
Community sanitation	−0.018 [−0.148, 0.112]	.784
WAZ		
Household water	−0.015 [−0.084, 0.055]	.677
Household sanitation	0.039 [−0.020, 0.098]	.191
Community water	0.081 [−0.021, 0.184]	.121
Community sanitation	0.022 [−0.092, 0.136]	.706
WHZ		
Household water	0.039 [−0.038, 0.117]	.318
Household sanitation	0.036 [−0.025, 0.096]	.244
Community water	0.019 [−0.099, 0.137]	.750
Community sanitation	0.045 [−0.070, 0.160]	.446

Note. N = 36,040 (HAZ), 36,308 (WAZ), 36,145 (WHZ); 4,426 clusters each.

One cell reaches significance at 5%: household water for HAZ, conditional on the other three terms (β = −0.085, p = .043), unexpectedly negative. Household and community water are correlated, and this is one significant cell among twelve in this table; it is reported as a single finding, not treated as central, and not read as evidence that improved household water is harmful.

5.4 Specification sensitivity: Nigeria community sanitation

Table 5.4 — Nigeria community sanitation–HAZ: specification-sensitivity results.

Specification	Sanitation–HAZ β	*p*
No controls	0.274	.004
+ child controls	0.296	.001
+ socioeconomic controls	−0.457	<.001
+ region FE	−0.184	.035
Household cluster FE (own-household sanitation)	−0.007	.919

The coefficient changes sign and grows in absolute size as controls are added sequentially. Removing wealth quintile alone from the full control set moves it from −0.457 back to −0.196, the largest movement from any single block — sensitivity to specification, not proof wealth drives the reversal. The household-level coefficient for Nigeria is close to zero and not significant: the community-level, region-FE estimate is **not stable across specifications**, not evidence that improved sanitation harms growth in Nigeria. Appendix B.5 re-estimates the same fully adjusted community model under the binary WHO cutoffs, an alternative operationalisation of the same anthropometric measurements rather than an independent sample: it finds three, not one, nominally significant cells among 30, including Nigeria's sanitation–stunted coefficient in the same direction as here.

5.5 Current-age heterogeneity

Six current-age bands (0–5 to 48–59 completed months, 12–23 as reference) replace the continuous age control (Section 4.4). Of twelve exposure-specific joint interaction tests (two exposures × three outcomes × two fixed-effects structures, χ^2, 5 restrictions each), eleven of twelve reject no heterogeneity at 5% (exception: water–WHZ under region FE, $p = .300$); the combined ten-term tests reject in all six combinations (largest $p = .019$). The household-WASH association with current outcomes therefore varies with current age in most specifications tested — secondary evidence about the *current* survey-date association only, which does not reconstruct any child's actual early-life exposure, the question the historical extension addresses under its own limitations.

5.6 Historical extension: primary established-date models

All cells come from the primary weighted models, post-12-month window, established-date sample (Ghana 2014, Kenya 2014, Nigeria 2018), per 10 percentage points of IHME-modelled local coverage — an area average within each child's buffer, not a household record (Section 3.5). These are weighted conditional associations among eligible measured children, not population-representative, with confidence intervals and p-values from the Student $t(G-1)$ distribution.

Table 5.5 — Historical extension: primary established-date water (W_IMP) estimates, with sample sizes.

Water (W_IMP): coefficients per 10 percentage points

Sample	β [95% CI]	*p*	N / G
HAZ			
Ghana 2014	1.65 [0.34, 2.96]	.014	N1,940 / G353
Kenya 2014	−0.61 [−2.07, 0.86]	.417	N14,166 / G1,498
Nigeria 2018	0.70 [0.15, 1.25]	.013	N6,912 / G1,263
Pooled established	**1.03 [0.51, 1.54]**	**<.001**	N23,018 / G3,114
WAZ			
Ghana 2014	1.06 [−0.16, 2.28]	.088	N1,940 / G353
Kenya 2014	−0.22 [−1.30, 0.86]	.690	N14,166 / G1,498
Nigeria 2018	0.45 [0.00, 0.90]	.050	N6,951 / G1,264
Pooled established	0.65 [0.20, 1.10]	.005	N23,057 / G3,115
WHZ			
Ghana 2014	−0.00 [−1.16, 1.16]	.997	N1,940 / G353
Kenya 2014	−0.16 [−1.29, 0.96]	.776	N14,166 / G1,498
Nigeria 2018	−0.05 [−0.48, 0.38]	.809	N6,942 / G1,264
Pooled established	−0.08 [−0.50, 0.35]	.717	N23,048 / G3,115

Note. N and G are children and clusters in each country/outcome's own estimation sample.

Table 5.6 — Historical extension: primary established-date sanitation (S_IMP) estimates.

Sanitation (S_IMP): coefficients per 10 percentage points

Sample	β [95% CI]	*p*
HAZ		
Ghana 2014	−2.48 [−4.91, −0.05]	.046
Kenya 2014	−0.21 [−0.53, 0.11]	.190
Nigeria 2018	0.20 [−0.34, 0.75]	.467
Pooled established	−0.10 [−0.37, 0.17]	.483
WAZ		
Ghana 2014	−0.08 [−2.04, 1.88]	.935
Kenya 2014	−0.15 [−0.38, 0.09]	.223

Table 5.6 (continued)

Sample	β [95% CI]	*p*
Nigeria 2018	0.02 [−0.44, 0.48]	.937
Pooled established	−0.07 [−0.29, 0.14]	.511
WHZ		
Ghana 2014	1.64 [−0.27, 3.56]	.092
Kenya 2014	0.00 [−0.22, 0.22]	.980
Nigeria 2018	−0.17 [−0.59, 0.24]	.407
Pooled established	−0.03 [−0.24, 0.17]	.742

Note. Sample sizes identical to Table 5.5 for each outcome/country; shown once there.

The pooled water–HAZ coefficient (1.03, $p < .001$) is this section's selected headline. It is **not** the only nominally significant cell: six of 24 reach $p < .05$ — pooled water–HAZ, pooled water–WAZ, Ghana water–HAZ, Nigeria water–HAZ, Nigeria water–WAZ ($p = .0498$) and Ghana sanitation–HAZ (negative). It was chosen as headline because it is the pooled estimate for this thesis's primary outcome and window, proposed in a dated specification document before any coefficient was estimated (Section 6.4), not because it is the only significant result. All 24 primary cells were independently reconstructed with a different estimation library on a separately rebuilt sample, matching stored values (Appendix B.2) — confirming the specification, correctly implemented, produces these numbers, not a multiplicity correction and not evidence of causality.

Weighted vs. unweighted. The pooled coefficient is sensitive to the equal-country weighting convention: 1.03 weighted versus 0.41 [0.03, 0.80], $p = .035$, unweighted on the identical 23,018 observations — a disclosed sensitivity (Section 4.5), not evidence either weighting is more "correct"; equal weight totals do not give equal influence on the pooled slope, which also depends on each country's residual exposure variation.

Residual exposure variation. The pooled model's residual within-cluster exposure standard deviation is 0.43 percentage points; the conventional 10-point coefficient scale is therefore approximately 23 times that standard deviation — a scale benchmark, not a single identifying contrast or a bound on observed changes. One residual SD corresponds to approximately 0.044 HAZ standard deviations.

5.7 Alternative windows and common-sample comparison

Table 5.7 — Historical extension: alternative exposure-window and common-sample comparison (pooled water–HAZ).

Window	Pooled water–HAZ β [95% CI]	Sample
Post-12-month (main)	1.03 [0.51, 1.54]	Own-eligible, N = 23,018
Birth-year	0.54 [0.12, 0.96]	Own-eligible
Prenatal (9 months)	0.64 [0.24, 1.03]	Own-eligible
Post-24-month	1.75 [0.81, 2.70]	Own-eligible
Post-12-month, common-sample	1.64 [0.68, 2.60]	All-window-eligible, N = 16,419

Note. Per 10 percentage points, as Table 5.5. "Own-eligible" is each window's own sample; the common sample is eligible under all four.

The first four rows each use their own window's eligible sample, so comparing them changes window and sample at once. The last row instead holds post-12-month fixed and changes only the sample, 1.03 to 1.64 — a **sample-composition effect**, not evidence favouring one window's validity. The table gives no like-for-like comparison of all four windows on one sample. No window was selected for its coefficient's significance; all four were listed together in the same dated pre-estimation specification (Appendix C).

5.8 Further sensitivity: wealth, spatial method, completion rule

Table 5.8 — Historical extension: further sensitivity of pooled water–HAZ.

Sensitivity	Pooled water–HAZ β [95% CI]
Main (post_12m, weighted)	1.03 [0.51, 1.54]
Wealth quintile added	1.01 [0.50, 1.52]
Containing-pixel assignment (vs. buffer average)	0.68 [0.26, 1.10]
Rural 10 km buffer (urban unchanged)	1.01 [0.42, 1.60]
Stricter completion rule (+1 month)	0.95 [0.43, 1.47]
Excluding Ghana (DHS/IHME input overlap)	0.74 [0.24, 1.24]

Note. Excluding Ghana pools Kenya and Nigeria only and is not overlap-free: Kenya's own DHS survey also feeds the IHME water product (Section 6.5).

Adding wealth leaves the coefficient essentially unchanged, showing only that it is not sensitive to this adjustment; it neither establishes mediation nor adjudicates the competing confounding concern (Section 3.8), nor bears on cluster-fixed-effects absorption (Section 4.2).

The spatial-method and completion-rule sensitivities move the coefficient by amounts smaller than, or comparable to, the weighted/unweighted gap; none is privileged over the main specification.

5.9 Ethiopia: provisional, outside the established-date sample

Ethiopia's three interview-date candidates are not equally valid. Candidate C1's 30-day Ethiopian-calendar arithmetic has already been shown invalid for the delivered date fields, independently of the WASH analysis (Appendix C); it is excluded from every reporting table. Candidates C2 and C3 remain provisional, neither independently validated; they agree closely (water–HAZ: −0.39 both; sanitation–HAZ: −0.45 both; all four confidence intervals include zero). These are reported only as an exploratory sensitivity, never pooled into or compared with the established-date headline results.

6 Discussion and limitations

6.1 Relating the findings to the literature

The baseline findings are consistent with the existing literature, but only narrowly. Headey and Palloni (2019), under a within-region panel design netting out time-invariant confounding, find sanitation improvements predict large reductions in diarrhoea and mortality but **not** stunting or wasting, while water is insignificant for most outcomes except piped water. Their null sanitation–stunting finding parallels this thesis's own null household- and community-sanitation associations with HAZ (Sections 5.1–5.2), each via its own design (region-level panel fixed effects there; region or cluster fixed effects here) — narrower than claiming WASH–growth associations generally attenuate under better controls, which would overstate what one study establishes.

The household cluster-fixed-effects associations (Table 5.1) are not significant at 5%, broadly consistent with the strongest randomised evidence available: no WASH-only arm in the three trials showed a detectable growth effect (Section 2.3), and adding WASH to nutrition brought no additional benefit over nutrition alone (Cumming et al. 2019). This is not confirmation — a facility-type classification in cross-sectional data and a randomised prospective intervention answer related but distinct questions. The one significant household-level result (Section 5.3) is negative and just below threshold, one cell among twelve, not evidence

improved household water harms growth; no mechanism in Section 2 supports that reading.

The historical extension's pooled water–HAZ association is positive and significant (Section 5.6), independently reproduced with a second estimation library on a separately rebuilt sample, as were all 24 primary cells (Appendix B). Computational reproduction is not a multiplicity correction: no formal adjustment has been applied across the baseline and historical analyses, so this thesis has neither quantified nor ruled out that this result is one significant draw among many tested cells. It would be tempting to read it as evidence that a time dimension recovers an association the single-round baseline cannot detect; Section 6.2 explains why this thesis does not draw that conclusion.

6.2 Why baseline and historical results cannot test exposure timing

This is the most tempting overinterpretation the thesis's structure invites. The historical extension's research question (Section 1.3) is framed to avoid it: it asks whether modelled early-life WASH coverage is associated with outcomes *within its own sample*, not whether assigning a historical rather than current exposure changes an otherwise identical estimate.

The baseline and historical samples differ in at least four respects at once:

1. **Children and survey rounds.** The baseline uses 2022–2025 rounds from four countries; the historical extension uses 2014–2018 rounds from three, Ethiopia entering only as a provisional sensitivity (Section 1.3). No child appears in both.
2. **Exposure definitions.** The baseline exposure is each household's own survey response, or an unweighted leave-one-out proportion among other households (Section 3.3); the historical exposure is a modelled, area-averaged annual gridded estimate matched to a postnatal window (Sections 3.5–3.7) — a different kind of measurement, not the same quantity at a different time.
3. **Eligibility rules.** The historical sample requires complete spatial coverage and a complete window verified against the interview date (Section 3.7), with no baseline counterpart.
4. **Control sets.** The baseline household model uses cluster fixed effects with socioeconomic controls (Section 3.4); region fixed effects appear in the region and community specifications but are never stacked with cluster fixed effects. The historical specification uses cluster fixed effects with country-by-birth-year terms, excludes wealth for its own documented reason (Sections 3.8, 4.6), and is identified by genuine within-cluster cohort variation; the baseline community model is deliberately not estimated with cluster fixed effects (Section 4.3), for a reason with no bearing on the historical design.

Because all four differ at once, the contrast between the baseline's largely null associations and the historical headline cannot be attributed to "the effect of assigning historical instead of current exposure" — that headline is one of six nominally significant cells among 24, not the only significant result (Section 5.6). The contrast could instead, or in addition, reflect different confounding structures, different measurement-error properties, the different control sets, or chance given the number of cells estimated. No formal test of the baseline-versus-historical difference was carried out. This thesis does not adjudicate between these explanations: its design, two separately constructed samples rather than one in which only timing varies, cannot do so.

The within-historical-sample window comparison (Section 5.7) holds constant the survey rounds, exposure source, control set, and estimator, changing only which months are averaged — but not the estimation sample, since each window has its own eligible children. With the window held at post-12-month, the estimate moves from 1.03 on its own eligible sample to 1.64 on the smaller common-eligible sample, a sample-composition effect, not evidence favouring one window. This comparison says nothing about the baseline-versus-historical contrast above.

6.3 Limitations: unresolved documentation

Two issues remain genuinely unresolved, neither a design choice nor an inherent limitation. First, no DHS or IPUMS source documents a sampling weight specific to the measured-anthropometry subsample (Appendix A); the working inference, not a documented fact, is that within-survey PERWEIGHT normalisation removes a constant selection factor under random household-level selection, though it cannot correct for differential non-response among eligible but unmeasured children. Second, the gap between Nigeria 2018's official measured-child count (12,806) and the IPUMS extract's non-missing-anthropometry count (11,704) remains unexplained (Section 3.9). Both bear on which children the historical estimates describe, so results are reported as **weighted conditional associations among eligible measured children**, not population-representative estimates; if measurement or non-response related to both WASH coverage and growth, the estimated associations could differ from the full eligible population in a direction the data cannot show.

6.4 Limitations: authorised design choices

Three features are deliberate, disclosed choices, not errors. First, equal-country-total weighting (Section 4.5) is not a population-size-weighted estimate or an average of separately estimated country coefficients — statistical influence also depends on residual exposure variation, so equal weight totals do not imply equal influence; the pooled water–HAZ coefficient moves from 1.03 (weighted) to 0.41 (unweighted), neither population-representative. Second, the working 12-month postnatal window was proposed as primary in a dated specification document before any model was estimated, because its completion can be checked against the interview date; that document explicitly left the final choice open for author/supervisor input (not resolved until later), so this thesis describes the window as proposed, not as "preregistered" or finally decided in advance. The literature basis is modest — Cumming and Cairncross (2016) lean toward a 24-month window — which is why all four candidate windows (Section 5.7) are reported together. Third, wealth quintile is excluded from the historical main specification because it may be a post-exposure mediator correlated with the exposure (Section 3.8), not a claim that cluster fixed effects remove wealth mechanically (Section 4.2 shows otherwise); the competing concern is that current wealth could instead proxy prior socioeconomic circumstances, left unaddressed if excluded. Adding wealth leaves the coefficient essentially unchanged (1.01 vs. 1.03; Section 5.8), without adjudicating between the two readings.

6.5 Limitations: inherent, not resolvable

DHS/IHME input overlap affects Ghana and Kenya, whose own DHS surveys feed the IHME estimates used as exposure; the overlap-exclusion sensitivity (pooled water–HAZ excluding Ghana: 0.74, Section 5.8) addresses only part of it, since Kenya's overlap remains. Current residence proxies early-life residence; children whose survey residence differs from their early-life residence may be assigned an exposure that misrepresents it, under both the buffer-average and containing-pixel methods — a measurement error of unknown extent. DHS cluster coordinates are randomly displaced for confidentiality; neither extraction method recovers the true location. The IHME surface is itself a modelled estimate with its own uncertainty not reflected in the reported standard errors, and its annual resolution means a window's final year can include months after interview even when the postnatal completion rule is satisfied (flagged, not excluded or adjusted for). The residual within-cluster exposure variation that identifies the historical coefficients is small (0.20–0.88 percentage points by country and product; 0.43 for the pooled water–HAZ model) relative to the 10-point reporting convention. Finally, Ethiopia's

interview-date uncertainty is inherent to that round's recoded date fields, not a design choice; its estimates remain provisional and outside the established-date pooled family.

6.6 What this thesis does not claim

This thesis does not claim its findings establish a causal effect of WASH facilities on child growth, in either direction: both analyses are observational, and fixed effects remove only what is shared within a region or cluster, leaving within-cluster confounding possible (Section 4.2). Nor does it read the null or weak baseline associations as evidence WASH investment is ineffective, generalise findings from these four countries to Sub-Saharan Africa (Section 1.3), treat computational verification (Appendix B) as evidence of improved causal identification, or treat the baseline/historical contrast as evidence that exposure timing explains it (Section 6.2).

7 Conclusion

This thesis examined the association between water and sanitation conditions and child anthropometric outcomes in four Sub-Saharan African countries: Ethiopia, Ghana, Kenya and Nigeria. Two distinctions ran throughout: household facilities versus surrounding community WASH, and a single survey-date exposure versus a preliminary, geospatially matched measure of early-life WASH coverage, built for an earlier survey round in three of the four countries, with Ethiopia retained only as a provisional sensitivity analysis (Section 1.3).

The baseline evidence shows no clear, robust association between improved household WASH and child outcomes within the same DHS cluster (Section 5.1): inconclusive, not evidence of no association. The community measure shows a large, significant association before covariate adjustment, attenuating to non-significance once socioeconomic status and region fixed effects are added (Section 5.2), consistent with, though not proof of, confounding by local development.

The historical extension's headline cell, the pooled established-date water–HAZ association (Section 5.6), is positive and significant; it carries the qualifications set out in full in Section 6 and cannot be read as evidence of an exposure-timing effect (Section 6.2). Neither analysis establishes a causal effect of WASH facilities on child outcomes, or a causal policy effect: the associations reported inform research and interpretation, not policy. The remaining gaps are detailed in Section 6 and Appendix C.

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Appendix A — Variable definitions, sample flow, and authoritative sources

A.1 Variable definitions

Baseline and historical variables are listed in separate tables, because the two samples are built by different source pipelines (country-specific DHS recode files for the baseline; IPUMS-DHS for the historical extension) and use different native field names for the same kind of quantity. A baseline field is never the same field as its historical counterpart under another name, even where the two serve an analogous purpose (Section 3.4, Section 3.8).

The unit of analysis in both samples is the child. Table A.1's sources are in `scripts/dhs_harmonization/`; Table A.2's are in `scripts/historical_wash/`. Each table cites its sources by a short code, keyed to the exact filename beneath that table; full paths and the complete script inventory are in the repository, not repeated per row here.

Table A.1 — Baseline variables (Sections 3.1–3.4)

Variable	Definition	Source
HAZ, WAZ, WHZ	Height-for-age, weight-for-age and weight-for-height z-scores, WHO Child Growth Standards reference (WHO Multicentre Growth Reference Study Group 2006). Read from the KR fields `hw70`, `hw71` and `hw72`, which store each z-score multiplied by 100; every valid value is divided by 100 once, giving the analysis scale in standard deviations. A value is valid unless it is system-missing or one of the DHS special codes 9996 (height out of plausible limits), 9997 (age in days out of plausible limits), 9998 (flagged cases) or 9999 (missing). All five non-valid states become missing, never a number. The bounds −6 to 6 (HAZ) and −6 to 5 (WAZ, WHZ) are a validation check that stops the pipeline if a valid value falls outside them; they do not reclassify any value.	KR; S1
Household water (binary)	1 if the household drinking-water source on the child's KR record (`v113`) is JMP-improved; 0 if it is unimproved or surface water. Code 96 (other; "Other / unclassified") and code 97 (not a de jure resident; "structurally not applicable") are set to missing, not to 0. "Improved" is a facility-type category; it is not a measure of a safe or safely managed service.	S2, S3, S4

Table A.1 (continued)

Variable	Definition	Source
Household sanitation (binary)	1 if the KR sanitation facility (`v116`) is JMP-improved, including an improved facility shared with other households (the Basic and Limited service levels derived from `v160` are both counted as improved); 0 if it is unimproved or open defecation. Code 96 ("Other / unclassified"), code 97 ("structurally not applicable"), an improved facility whose sharing code is 7 (also structurally not applicable) and Ghana's flush-to-biodigester code 16 (unclassified in the core specification) are set to missing.	S2, S3, S4
Community water/sanitation (`water_rate_loo`, `sanitation_rate_loo_core`)	Unweighted leave-one-out proportion of improved facilities among the *other* sampled, classifiable Household Recode (HR) households in the child's DHS cluster, classified from `hv201`, `hv205` and `hv225` with the same mapping as the household variables. The child's own household is excluded from both the numerator and the denominator, so all children in one household receive the same value. Households recorded as Other / unclassified or structurally not applicable, and, in the core sanitation variant, households using Ghana's unclassified biodigester, are also excluded from both. The value lies on the 0–1 scale and is missing when no other classifiable household remains in the cluster. HR covers sampled households only, so the measure is not a census of every household in the physical community, and it is not an average over children. A robustness variant, `sanitation_rate_loo_robustness_biodigester`, counts code 16 as improved; it differs from the core variant only in Ghana.	HR; S5, S2
`v005` → `weight_original`	The woman's individual sample weight from the DHS Children's Recode (KR), divided by the documented factor of 1,000,000. Each child's KR record carries the mother's weight, because DHS computes no separate per-child weight within KR. The distinct Births Recode (BR) is not used. This is **not** the household weight `hv005`, which exists only in the Household Recode (HR) and is not used in the baseline models. Pooled models rescale `weight_original` to equal country totals within each model's estimation sample (Section 4.5); country-specific models use it unchanged.	S6; rescaling in S7

Table A.1 (continued)

Variable	Definition	Source
`v021` → `psu` → `country_psu`	DHS's native primary sampling unit code, copied to `psu` and combined with a country prefix into the country-qualified cluster key `country_psu`. Pooled models use `country_psu` for cluster fixed effects and clustered standard errors; country-specific models cluster on `psu`.	S6, S8
`b19` (child's own age in months)	The DHS recode's standard current-age-in-months field, stored unchanged as `b19_raw` and used uniformly for all four baseline countries (no country-varying construction); a linear control from Model 2 onwards.	S6, S7

Note. Source key: S1 `02_load_flag_anthro.py`; S2 `01_wash_mapping.py`; S3 `03_build_indicators.py`; S4 `12_robustness.py`; S5 `08_hr_cluster_wash_exposure.py`; S6 `04_build_country_kr.py`; S7 `11_main_regressions.py`; S8 `05_pool_countries.py` (all in `scripts/dhs_harmonization/`). KR/HR: DHS Children's/Household Recode, Ethiopia 2024–25, Ghana 2022, Kenya 2022, Nigeria 2024; the baseline does not use IPUMS-DHS. The pipeline applies the same special-code list to all four countries; whether DHS's own criteria behind code 9998 are identical across these surveys is not established here. Community exposure groups HR households by the HR cluster number `hv001` and links them to children through `v001` and `v002`, whereas cluster fixed effects and clustered standard errors use `v021`. Whether these identifiers correspond one-to-one in every baseline country has not been checked against the restricted data and remains an open verification question.

Table A.2 — Historical-extension variables (Sections 3.5–3.8)

Variable	Definition	Source
`IDHSPSU`	Sample-qualified DHS cluster code from IPUMS-DHS, unique across survey rounds; used for cluster fixed effects and clustered standard errors. **It is the historical extension's own cluster identifier, distinct from the baseline's `country_psu` above.**	H1
`PERWEIGHT`	The IPUMS-DHS standard sampling weight for child-level tabulations, rescaled to equal country totals within each model's final estimation sample (Section 4.5). **It is the historical extension's own weight field, distinct from the baseline's `weight_original` above.** It is not a documented anthropometry-specific weight (Section A.4). `KIDWT`, an IPUMS population adjustment factor, is not used.	H1

Table A.2 (continued)

Variable	Definition	Source
W_IMP, S_IMP	IHME modelled annual local coverage of JMP-improved water and sanitation: percent of the population with access, on a 0–100 scale, ~5×5 km grid, 2000–2017. W_IMP sums piped water (on- or off-premises) and other JMP-improved sources (protected wells and springs, bottled water, rainwater collection, bought water); S_IMP sums sewer/septic sanitation and other JMP-improved facilities (improved and ventilated improved latrines, composting toilets); both verified directly from the primary source, not inferred from the raster scale or file names. Each child's value is the area average of the MEAN surface within a buffer of 2 km (urban) or 5 km (rural) around the displaced cluster coordinates; it is not population-weighted. Containing-pixel assignment and a rural 10 km buffer are sensitivity methods.	H3
`age_m` (child's own age in months)	**The historical extension's own age field, distinct from the baseline's `b19` above.** Reported completed months (`KIDCURAGEMO`) where the source delivers them (Ethiopia, Nigeria); otherwise the documented calendar-month difference between interview and birth dates (Ghana, Kenya).	H1, H2
`b_year` (birth year)	The child's birth year, from the birth-date century-month code; used for survey-qualified country-by-birth-year terms in the historical models. Derived in the analysis-sample construction stage and confirmed independent of the Ethiopian interview-date candidate (Appendix C).	H2
`cluster_key`	A copy of `IDHSPSU` (`cluster_key = IDHSPSU`), which already encodes survey and cluster, so the key is sample-qualified by construction.	H2; RESTRICTED analysis-candidates store

Note. Source key: H1 `16_estimate_preliminary.py`; H2 `14_build_analysis_samples.py`; H3 `10_spatial_extraction_v3.py`; H4 `12_construct_exposure_windows.py` (all in `scripts/historical_wash/`; W_IMP/S_IMP product definitions also recorded in `docs/provenance/historical_wash_ihme_product_definitions_v1_2026-10-07.md`). Historical source: the IPUMS-DHS child-level extract for Ghana 2014, Kenya 2014 and Nigeria 2018 (the established-date family); Ethiopia 2016 enters only as a provisional sensitivity analysis (Appendix C). Historical HAZ, WAZ and WHZ come from the IPUMS fields `HWHAZWHO`, `HWWAZWHO` and `HWWHZWHO` (stored multiplied by 100 and divided by 100 once); code 9999 marks a child outside the measurement universe, and codes 9995–9998 and system-missing values are not valid. IHME raster values are percentages; NoData pixels are excluded from the valid area rather than treated as zero. A child-year counts as complete only when the valid area covers the buffer to within 10^{-3}; partial-coverage values are kept for diagnostics only. The supplied LOWER and UPPER surfaces are not regression confidence intervals and do not enter the models. Exposure windows (H4): the birth-year window is the full calendar year of birth; the prenatal window is the nine months before the birth month (birth month excluded); the post-12- and post-24-month windows start with the birth month (included). Each calendar year is weighted by its share of the window's months, and a window needing any year outside 2000–2017 is not assigned a value. Only the two postnatal windows require every window month to have ended before the interview month;

the prenatal and birth-year windows have no interview-date completion check. Because the surface is annual, a value may include months after the interview; such cases are flagged, not excluded (Section 3.7).

A.2 Baseline sample flow

Counts are unweighted numbers of children.

1. Pooled harmonised child file: 70,231 children, 4,481 clusters and 114 regions.
2. Outcome-valid samples within the measurement universe: 36,985 children for HAZ, 37,260 for WAZ and 37,088 for WHZ, in 4,434 clusters. Validity is the outcome rule in Table A.1; the code applies no separate measurement-universe filter. No child is lost at the exposure or covariate step.
3. Common samples requiring both household and community exposures to be classifiable: 36,040 (HAZ), 36,308 (WAZ) and 36,145 (WHZ) children, in 4,426 clusters and 25,627, 25,759 and 25,665 households.

The baseline code contains no singleton-cluster exclusion step (Section 4.5), so no baseline singleton count exists. Full flow: `outputs/household_community_wash/01_development_sample_flow.csv` and, for the outcome-valid samples, `outputs/tables/step11_model_sample_flow.csv`.

A.3 Historical sample flow (established-date, post-12-month window, water, HAZ, as a worked example)

The example is the primary pooled model (Ghana 2014, Kenya 2014, Nigeria 2018; W_IMP; primary buffer method; weighted).

1. Raw analysis candidates in the three established-date samples: 60,772 child records (Ghana 5,884; Kenya 20,964; Nigeria 33,924).
2. Candidate label `GREGORIAN_ESTABLISHED`: 60,772 (no further loss for the three established countries).
3. Complete postnatal window and complete spatial coverage: 42,052.
4. HAZ valid and within the measured universe: 23,135.
5. Complete cases on covariates, exposure, weight and cluster key: 23,135 (no further loss).
6. Singleton-cluster exclusion: −117 clusters and −117 observations.
7. **Final: 23,018 children in 3,114 clusters** (Ghana 1,940 in 353; Kenya 14,166 in 1,498; Nigeria 6,912 in 1,263).

Singleton clusters are removed separately from each model's own final sample, so the number removed differs between models; the three country models remove 27, 16 and 74. Section 3.9 explains why the Nigeria counts differ from the official 2018 anthropometry count; that gap remains unresolved. Steps 1, 2, 4, 6 and 7 are checked against public aggregate files in `docs/provenance/`; step 3 is recorded only in the restricted log, and was independently reproduced in a separate validation run (RESTRICTED execution log).

A.4 The measurement-subsample weight: an unresolved documentation gap (Section 3.8, Section 6.3)

No DHS or IPUMS source consulted for this thesis documents a sampling weight specific to the anthropometry-measured subsample, as distinct from the general household/individual sampling weight. Both DHS's own Biomarker Manual and Guide to DHS Statistics were unreachable (HTTP 403) at the time; the accessible sources checked instead (the World Bank's microdata catalogue, the Nigeria 2018 NDHS final report, and the IPUMS-DHS PERWEIGHT/KIDWT documentation) confirm what weight fields exist but give no measurement-subsample-specific guidance. This is recorded as an unresolved documentation gap, not a problem this thesis attempts to resolve by substitution: no replacement weight or correction was invented, and Section 3.8 states the working inference used instead. The sources tried and their individual results are kept in the repository's own issue register.

A.5 Full authoritative-file index

The current index of baseline and historical output files cited in this thesis, with their status labels (validated for main reporting, validated sensitivity, exploratory, invalid or superseded, not estimable), is kept in the repository's `docs/provenance/` directory, cross-referenced by result rather than reproduced here.

A.6 Descriptive figures

The baseline sample's descriptive anthropometric-prevalence and household-WASH-characteristics figures are reported in the main text, next to the variable constructions and sample descriptions they illustrate (Figure 3.2, Section 3.2; Figure 3.3, Section 3.3), not duplicated here.

Appendix B — Model specifications, inference conventions, and independent verification

The estimating equations are stated in Chapter 4 (Sections 4.1–4.6). This appendix records how they are implemented for inference and how far the implementation has been verified: the baseline household cluster-fixed-effects model (Section B.1), the historical extension's independent re-estimation (Section B.2), the design checks applied to all 24 primary historical models (Section B.3), and an estimator safeguard added after an external audit (Section B.4). Figures marked RESTRICTED come from execution logs and output folders that contain or derive from restricted microdata; the full execution logs, diagnostic scripts, and byte-for-byte comparison records underlying every verification summarised in this appendix are preserved in the project's own `docs/provenance/` directory and are not reproduced step by step here.

B.1 Inference-convention comparison (baseline household cluster-FE model, HAZ)

Table B.1 — Household water and sanitation coefficients for HAZ under the two inference conventions (baseline household cluster-fixed-effects model)

Convention	Water β	SE	p	Sanitation β	SE	p
PRIMARY (same-group CRV1; this thesis's main convention)	−0.085	0.046	.062	0.026	0.043	.548
FULL_DUMMY_DF sensitivity	−0.085	0.049	.080	0.026	0.046	.574

Note: Household water and sanitation are entered together; N = 36,040 children in G = 4,426 PSUs (`country_psu`). Both rows use the normal reference distribution for *p*-values and confidence intervals. Source: `outputs/household_community_wash/03_household_cluster_fe.csv`, rows `natural_sample_PRIMARY_SAME_CLUSTER_CRV1` and `natural_sample_FULL_DUMMY_DF_SENSITIVITY`, outcome `haz`.

The two conventions differ only in whether the small-sample CRV1 correction also penalises degrees of freedom for the absorbed cluster dummies. FULL_DUMMY_DF does; PRIMARY does not, following the project's cross-validated adjudication against `linearmodels.PanelOLS(auto_df=True)` (Sheppard 2025) (`docs/provenance/cluster_fe_inference_adjudication.md`). Only standard errors, confidence intervals and *p*-values differ; the point estimates remain unchanged. The

adjudicated PRIMARY convention is the one presented in the abstract, the introduction and the conclusion; the FULL_DUMMY_DF sensitivity is reported next to it in Section 4.5 and Section 5.1. The correction is $c = \frac{G}{G-1} \times \frac{N-1}{N-K}$, where PRIMARY counts $K = 18$ demeaned regressors (no constant, no PSU dummies) and FULL_DUMMY_DF counts every absorbed PSU dummy as an estimated parameter ($K = 18 + 4{,}426$), multiplying the PRIMARY standard error by $\sqrt{(N-18)/(N-4{,}444)}$ (1.0677 for HAZ). The baseline code applies no singleton-cluster exclusion, so G counts every PSU in the estimation sample (Section 4.5); the region-fixed-effects models instead use explicit region dummies and the normal reference distribution throughout, and are not part of Table B.1. The baseline pipeline's own executed run log records `use_t: False`; independent recomputation confirms its stored p-values match a normal (z) reference distribution, not $t(G-1)$, to more than ten decimal places (e.g. the household water–HAZ cell's two candidate values, 0.0617391913770433 vs. 0.0618052, both round to the displayed .062, which is why a rounded-table check alone would miss the difference). A secondary cross-check against `linearmodels.PanelOLS` reproduces the stored PRIMARY standard errors once the library's `group_debias` adjustment is matched to the native estimator's own convention; the stored, published standard errors in Table B.1 come from `statsmodels`, not from this or any `linearmodels` cross-validation.

The historical and baseline pooled-weighting conventions differ only by a reporting-level constant (Section 4.5). Both give every country an equal total weight; the historical pipeline additionally multiplies every weight by N/M (N = estimation-sample size, M = number of countries), so each country's weights sum to N/M rather than 1. Because N/M is the same constant for every observation in a given model, and weighted least squares with its cluster-robust sandwich variance is invariant to rescaling every weight by one common constant, this changes no reported coefficient, standard error, confidence interval, or p-value — confirmed directly by refitting under both conventions, which reproduces identical coefficients and SEs to machine precision.

B.2 Independent verification of the pooled historical model (Section 4.6, Section 5.6)

This thesis's own pipeline (`scripts/historical_wash/16_estimate_preliminary.py`) estimates all 291 historical models with a hand-written estimator based on weighted within-cluster demeaning. An independent check re-estimated four representative country-specific cells (Ghana water–HAZ, weighted and unweighted; Ghana

sanitation–WAZ; Nigeria sanitation–HAZ) using a separately rebuilt sample and an explicit-dummy CRV1 implementation: coefficients agreed within 3×10^{-14}, standard errors within a relative 7.5×10^{-10}, and N and G were identical in all four cells.

That check did not cover the pooled multi-country model, the headline result of the historical extension. A separate, independent re-estimation rebuilt the estimation sample directly from the analysis-candidates stage (reproduced in Appendix A, Section A.3) and estimated it with `linearmodels.PanelOLS`, a different library from the project's own hand-written estimator, using cluster fixed effects and clustered CRV1 standard errors, under both inference conventions and both the weighted and unweighted specification. The rebuild is independent in filtering, weighting, design construction and estimation, but reads the same analysis-candidates file as the main pipeline, so it does not re-verify the variable constructions (exposure assignment, outcome validity, birth year, calendar age, weight field) built upstream of that file.

Table B.2 — Independent re-estimation of the pooled established-date water–HAZ model compared with the stored estimate

Quantity	Independent (corrected)	Stored (`m0004`)	Absolute difference
β per 10 points	1.0255531	1.025553067	6.7×10^{-8} against the six-decimal value 1.025553 (see note)
SE per 10 points (as originally run)	0.260694	0.260730	3.6×10^{-5} (explained below)
N	23,018	23,018	0
G (clusters)	3,114	3,114	0
K (regressors) / rank	24 / 24	24 / 24	0
Residual exposure SD (points)	0.428348	0.428348	none at six decimals
Unweighted counterpart β per 10	0.4147	0.4146971 (`m0008`)	none at four decimals

Note: Stored values are from `docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv` (rows `m0004` and `m0008`; β per 10 for `m0004` is 1.025553067200643 at full precision). The independent script compared its estimate with six-decimal constants written into its own code, so the 6.7×10^{-8} in the β row measures that rounding, not a disagreement between estimators; a full-precision comparison for this cell gives 2.4×10^{-14}. The independent rank is that of the regressor matrix before the fixed-effect transformation; the stored rank is that of the weighted within-transformed matrix. Independent values are carried from a RESTRICTED execution log; where only rounded values are recorded, agreement is stated at that precision.

An initial run of the independent script used a single pooled birth-year reference cell (rather than a country-specific reference year) and omitted the child's own age in months as a control, producing β per 10 points = 1.385 against the stored 1.026 — a roughly 35% discrepancy, despite the sample size, cluster count, and exposure mean/SD matching the stored values exactly.

A dedicated diagnostic traced this gap by decomposing the two changes on the transformed (weighted, within-cluster-demeaned) design the estimator actually uses, the space in which the exposure coefficient is identified. The two birth-year reference schemes were shown to be equivalent nuisance specifications for the exposure coefficient: both demeaned birth-year blocks span equal subspaces, the exposure's own residual after each full nuisance specification is identical at floating-point precision ($\leq 2.2 \times 10^{-16}$ relative), and the mechanism is that Nigeria's birth-year dummies under the single-reference scheme sum to its country indicator, which the cluster fixed effect absorbs regardless of scheme. The age-in-months control, not birth-year coding, accounts for the full gap: including it reproduces the stored coefficient (1.0255530672006674) under either birth-year scheme, and excluding it reproduces the original attempt's value (1.3852855197991372) under either scheme — differences of one to three units in the last place of a double-precision number. This is the diagnostic's finding on a reconstruction of the described design; it is not a claim about what the specific, unpreserved original code did, and not an exact recovery of that code's reported output. The complete five-step decomposition, tolerances, and execution log are in `docs/provenance/` and the diagnostic script's own docstring (RESTRICTED execution log; the values above are reproduced from it).

The SE difference in Table B.2 is a validation-script covariance-setting discrepancy, not a flaw in the published numbers. The pipeline's declared small-sample correction is $\frac{G}{G-1} \times \frac{N-1}{N-K}$, applied to the raw cluster-robust sandwich, with confidence intervals and p-values from a Student $t(G-1)$ distribution; K counts only the non-fixed-effect regressors (24 for the pooled cell). The independent script's original `PanelOLS` call applied only one of the two multiplicative factors this requires (`auto_df=True`, which correctly sets the degrees-of-freedom convention, but not the separate, opt-in `group_debias=True`, which applies the library's cluster-count adjustment). Adding `group_debias=True` and computing confidence intervals and p-values explicitly from the corrected SE with Student $t(G-1)$ — rather than `linearmodels`' own native reference, which counts every absorbed entity effect in its degrees of freedom and differs by up to 0.0065 in a CI bound in the smallest-cluster-count cells — reconciles all 24 primary cells to the stored values at better than 10^{-9} precision. None of this changes the stored, authoritative standard errors in the output of the pipeline's own estimator; what was corrected is the covariance configuration of the independent validation script, not the pipeline being validated. The SE in Table B.2 is kept as originally run.

Coverage across the 291 historical models. The verification described above covers the following, and no more: all 24 primary established-date cells (2 exposures × 3 outcomes

× Ghana, Kenya, Nigeria and pooled) were re-estimated with a second library on separately rebuilt samples; two further unweighted-counterpart cells were re-estimated individually; the remaining 265 models were not re-estimated with a second implementation, and were checked only by the estimator's own internal safeguards (Section B.4).

What this verification establishes, and what it does not. It establishes that the reported pooled coefficient and its standard error are computationally correct: they are the numbers the documented specification produces when correctly implemented in a second, independent library with an equivalent covariance configuration. It does not establish that the specification identifies a causal effect, that the samples represent the surveyed countries' populations, that the weighting and selection limitations are resolved, or that the result is robust to the number of cells tested (no multiplicity adjustment is applied). Nor does it place this cell in a higher reporting tier than its sibling historical models, which share the same unresolved documentation issues (Section 6; `docs/provenance/historical_wash_issue_register_v1_2026-10-07.md`).

B.3 Design checks applied to all 24 primary historical models

The following were checked programmatically for all 24 primary models: record-key uniqueness; the cluster-to-survey mapping (one survey per cluster); country weight totals equal to N/M within each model's own estimation sample (relative tolerance 10^{-6}); full design-matrix rank (the rank recorded by the estimator equals the number of retained regressors); and the absence of singleton clusters in the final estimation data. All checks pass. The per-model results are public (`docs/provenance/historical_wash_preliminary_estimates_v1/design_checks_per_model_PRELIMINARY_AGGREGATE.csv`, 24 rows); the summary file is RESTRICTED.

M is the number of countries in that particular model: $M = 1$ for the 18 country-specific models and $M = 3$ for the 6 established-date pooled models, not $M = 3$ for all 24. The historical scripts set each country's weight total to N/M; the baseline pooled models set each country's total to 1 (Section 4.5). The two differ by the common constant N/M, which changes neither the weighted least-squares coefficients nor the cluster-robust variance, since multiplying every weight by the same constant scales the sandwich's bread by its inverse and its meat by its square, and the two cancel. Equal country weight totals do not mean equal influence on the pooled slope, which depends on each country's residual exposure variation (Section 4.5).

B.4 Estimator safeguard: the exposure-absorption and rank-deficiency defect, and its resolution (2026-10-10)

An external independent audit identified a genuine code defect in the historical pipeline's estimator (`docs/provenance/independent_audit_resolution_v1_2026-10-10.md`, finding R05): the absorbed-column filter dropped any column with zero within-cluster variation after demeaning, but the exposure coefficient and its variance were then read by fixed array position rather than by the retained variable's name. Had the exposure itself ever been absorbed by this filter, the next retained column would silently have occupied its position, and its coefficient would have been reported as if it were the exposure's. Separately, the function computed the design-matrix rank but never compared it with the retained column count before solving by pseudoinverse, so an accidental exact collinearity would have returned a silently under-identified estimate rather than an error.

Fix. The estimator now refuses to proceed (rather than silently misreporting) whenever the exposure variable does not survive the absorbed-column filter, or whenever the retained design matrix is rank-deficient; when the exposure does survive, its coefficient and variance are retrieved by name, not by position. Neither change alters the computation for an estimable, full-rank case, since the exposure is always constructed as the first design-matrix column and so still occupies position 0 whenever it is retained. Four synthetic test cases (an ordinary estimable exposure; an exposure absorbed within every cluster; an exposure made exactly collinear with a retained control; a directly rank-deficient design) confirm the safeguard triggers exactly where intended and leaves the ordinary case unaffected.

Verification that no published model was affected. The patched estimator was re-run in full against the actual restricted inputs (confirmed by SHA-256 to be bit-identical to the inputs of the original stored run), writing a fresh 291-model output to a new run folder and leaving the original run's output unchanged. A machine comparison of the two runs found all 291 model IDs present in both with no duplicates, identical specification fields and `estimated` status in every row, and a maximum absolute difference of exactly 0.0 across all 15 numeric columns checked (coefficient, SE, both CI bounds at both scales, *p*-value, exposure variation diagnostics, N, G, K, rank); the two output files are byte-for-byte identical. This confirms directly, not by inference from the design checks alone, that none of the 291 published historical models was affected by this defect: in every one, the exposure was retained and the design matrix had full rank.

B.5 Binary-outcome robustness (linear probability models)

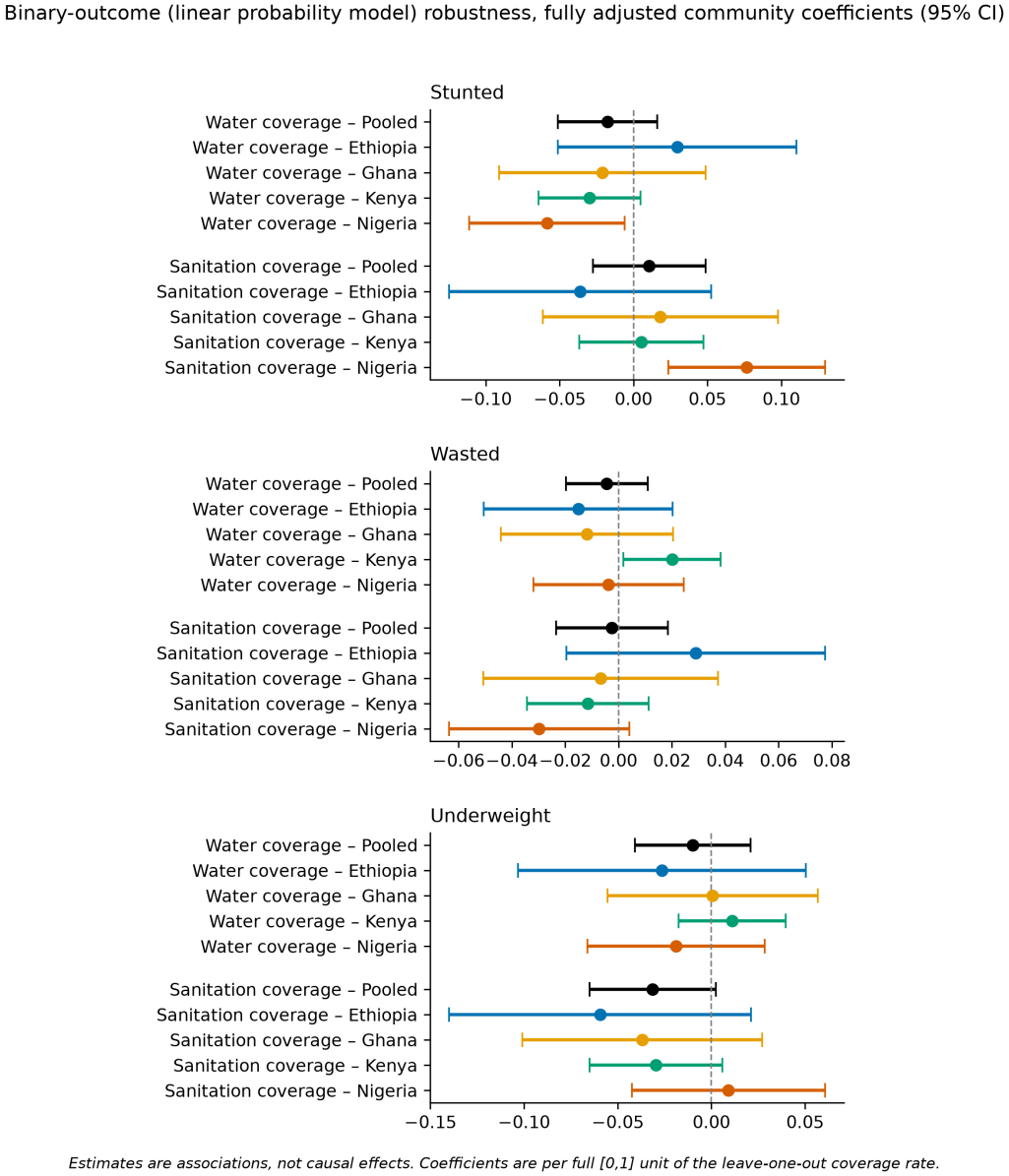


Figure B.1 — Fully adjusted community water and sanitation coefficients under the binary WHO stunting/wasting/underweight cutoffs (linear probability models), pooled and by country, with 95% confidence intervals.

The baseline community model (Table 5.2's fully adjusted stage, Figure 5.1) is re-estimated with the standard WHO binary indicators (stunted, wasted, underweight: z-score below −2) in place of the continuous HAZ/WAZ/WHZ z-scores, as a linear probability model under the same region fixed effects, controls, weighting and clustering. This checks whether the continuous-outcome pattern is an artefact of the z-score scale rather than the facility classification. Sample, weights and clustering are unchanged from the corresponding continuous-outcome model

(e.g. Nigeria, N = 9,410, G = 1,343, `cluster_robust_psu`). Coefficients are linear-probability differences per full unit of the leave-one-out coverage rate, which is a [0, 1] proportion (Section 3.3), not per 10 percentage points as in the historical tables (Section 5.6); they are not odds ratios, and predicted probabilities are not constrained to [0, 1].

Three of the 30 country–outcome–exposure cells shown reach $p < .05$, unadjusted for multiple comparisons: Nigeria's community sanitation coefficient on the stunted indicator (0.077, 95% CI [0.024, 0.130], $p = .005$), Nigeria's community water coefficient on the same indicator (−0.059, 95% CI [−0.111, −0.006], $p = .029$), and Kenya's community water coefficient on the wasted indicator (0.020, 95% CI [0.002, 0.038], $p = .031$). The Nigeria sanitation cell points the same direction as the continuous model's negative HAZ coefficient (Section 5.4): higher community sanitation coverage is associated with a higher probability of stunting as well as a lower mean HAZ. The Nigeria water and Kenya water cells have no counterpart flagged as significant in the continuous-outcome, country-specific results (Figure 5.1) and are reported here without further interpretation. These are alternative operationalisations of the same underlying anthropometric measurements, not an independent replication sample; with three nominally significant cells among 30 and no formal multiplicity adjustment applied, the number expected by chance alone is not negligible. The full 30-cell coefficient table is `outputs/final_appendix/appendix_A6_full_binary_lpm_coefficients.csv`.

Appendix C — Historical exposure-assignment diagnostics and the Ethiopia provisional analysis

This appendix holds exploratory material and is consistent with Section 5.9. Nothing in it is part of the established-date pooled headline result, and none of it should be cited as such. It has two parts. Sections C.1–C.3 cover Ethiopia 2016, which is analysed only as a separately labelled provisional sensitivity outside the established-date pooled sample of Ghana 2014, Kenya 2014 and Nigeria 2018. Section C.4 summarises the spatial-extraction rule and selected sensitivity results of the established-date family.

The reporting status of each part follows the final reporting map (`docs/provenance/final_empirical_package_v3_2026-10-07/01_final_reporting_map.md`). Ethiopia candidate C1 is **invalid and superseded**. Candidates C2 and C3 are **provisional and exploratory**. The rows of Table C.3 belong to the **sensitivity family** of the established-date models. Like every historical estimate in this thesis, they are conditional associations among eligible measured children, not causal effects and not population estimates.

C.1 Ethiopia's three interview-date candidates

For Ethiopia 2016, the IPUMS-DHS extract provides no Gregorian interview date. It provides the interview year, month and day in the Ethiopian calendar and a century-day code for the interview date (`docs/provenance/historical_wash_milestone_2026-10-06.md`, section 4). The child's birth date is a different field: the Gregorian birth century-month code that IPUMS supplies already converted, at month precision (`KIDDOBCMC`). All three candidates share that birth date. The interview-date candidates therefore differ only in the interview date.

The interview date and the birth date enter the historical construction at different points (`scripts/historical_wash/12_construct_exposure_windows.py`, `14_build_analysis_samples.py`, `16_estimate_preliminary.py`):

1. **Birth date.** Each window's months, and hence its annual weights and its exposure *value*, are placed from the birth century-month code alone. The birth year, and with it the survey-qualified birth-year terms, is identical under all three candidates for all 10,641 Ethiopian children.
2. **Interview date.** The candidate interview date decides whether a child's dates are valid at all and whether the birth precedes the interview; both conditions apply to every

window. It also decides whether a postnatal window has been completed by the interview (the post-12-month and post-24-month windows only), and it sets the descriptive flag for an annual value that covers months after the interview.

3. **Age.** The under-five rule and the age control in the Ethiopia models use the reported completed age in months (`KIDCURAGEMO`), which the extract supplies for all 10,641 Ethiopian children (`docs/provenance/historical_wash_analysis_samples_v1/age_rule_reconciliation_AGGREGATE.csv`). Neither depends on the candidate. A calendar age computed from the candidate interview date exists in the data but is used only in the calendar-age sensitivity, which is estimated for the established-date samples and not for Ethiopia.

The three candidate conversions of the interview date are (`scripts/historical_wash/06_ethiopia_calendar_audit.py`):

- **C1** ("triple": 30-day Ethiopian-calendar arithmetic applied directly to the delivered year, month and day fields). **Invalid and superseded.** The delivered Ethiopian day and month fields are not original 30-day Ethiopian dates: the day value 31 occurs in months that the Ethiopian calendar defines as having 30 days. Direct 30-day arithmetic is therefore invalid for these fields. The milestone record stating this was committed on 6 October 2026, before the historical outcome models were estimated; the finding does not depend on any outcome result. C1 was nevertheless estimated as a provisional candidate, then reclassified as invalid and superseded on 7 October 2026 (`docs/provenance/historical_wash_ethiopia_candidate_status_v1_2026-10-07.md`, section 1). Under this method, 275 of the 10,641 Ethiopian children have a day value that cannot be converted at all, so C1 gives them no date; the conversions for the other children use the same invalid logic. C1 is excluded from every reporting table in this thesis. Its earlier numerical agreement with C2 and C3 is not treated as evidence for either of them, because a method already known to be invalid cannot corroborate another one.
- **C2** ("century-day route"): the delivered century-day code read with the origin 1 Meskerem 1900 = 12 September 1907. Provisional; not independently validated.
- **C3** ("century-day route", anchored): 1 Meskerem 2008 = 12 September 2015, with the delivered month and day fields counted in Gregorian cumulative month lengths. Provisional; not independently validated.

The candidate-status record (section 2) names the weakness that C2 and C3 share, quoting the milestone record: the century-day code follows a month-length structure that

differs from a 30-day count, and its origin and the source of a constant two-day offset are not established. Neither candidate has been independently validated.

C.2 How closely C2 and C3 agree on window eligibility

Table C.1 — Ethiopia 2016: children eligible for each exposure window under candidates C2 and C3

Window	C2 eligible	C3 eligible	Disagree
Birth-year	9,862	9,862	0
Prenatal (9 months)	9,862	9,862	0
Post-12-month	7,930	7,922	8
Post-24-month	5,974	5,969	5

Note. Counts of children, out of 10,641, whose exposure for the window is eligible under the primary complete-coverage construction, before any outcome-specific exclusion; they are not model samples. "Disagree" counts children who are eligible under one candidate and not under the other. Each row is a different window for largely the same children, so the rows must not be added. All four postnatal counts (C2/C3, post-12/post-24) were computed directly from the restricted analysis-candidates file for this revision (filtering to each candidate and counting `eligible_complete` status) and match exactly. In both postnatal rows the difference between the two counts equals the number of disagreements.

Table C.1 reflects how the interview date enters eligibility (Section C.1). In the birth-year and prenatal windows, the interview date matters only through date validity and the requirement that birth precede the interview, and C2 and C3 classify every child in the same way. In the two postnatal windows, the completion check compares the window's end with the interview month, and the two candidates place a few borderline children on different sides of it: 8 children in the post-12-month window and 5 in the post-24-month window. The stored analysis samples show the same pattern. For water and HAZ, the birth-year and prenatal samples are 8,250 children under both candidates, while the post-12-month samples are 6,580 under C2 and 6,573 under C3 before singleton clusters are dropped, and 6,568 and 6,561 in the estimated models (`docs/provenance/historical_wash_analysis_samples_v1/design_diagnostics_by_sample_window_AGGREGATE.csv`; `model_coefficients_PRELIMINARY_AGGREGATE.csv`, rows `m0045` and `m0046`).

The comparison with C1 is narrower. For the birth-year window, the stored aggregate shows that C1 differs from C2 and C3 for exactly 275 children, all of them the children without a C1 date; no candidate places any birth after the interview (`docs/provenance/historical_wash_analysis_samples_v1/ethiopia_`

`birthyear_attribution_AGGREGATE.json`). This stored comparison covers the birth-year window only.

These counts support two narrow statements. First, the choice between C2 and C3 changes the window classification of very few children. Second, this is **not** evidence that either candidate's absolute date is correct. The two candidates share the unexplained recode structure described in Section C.1, so their agreement is not independent validation. Nor is the difference between them a confidence interval, an uncertainty bound or a guaranteed range for the true interview date: both could be wrong in the same direction.

A valid Gregorian interview date is still required for every child in every window, because it decides whether the child is validly dated and whether the birth precedes the interview (Section 3.7). This requirement is distinct from the postnatal-window **completion** check, as Section 3.7 and the construction script state. The completion check asks whether all months of the window had ended by the interview month, and it applies only to the post-12-month and post-24-month windows. The prenatal window ends before birth and needs no such check. The birth-year window has no completion check at all: a child born early in a year and interviewed before that year ends is still assigned that year's full annual value. Eligibility in the birth-year window still requires valid dates, a birth year inside the 2000–2017 raster span and complete spatial coverage. In any window, the annual value used for the interview's own calendar year can cover months after the interview; such cases are flagged and not excluded (Section 3.7). The candidate-status record (section 4) describes the completion rule as applying to every window, including the birth year, citing the design draft; the implemented construction does not do this, and Section 3.7 and this appendix follow the implementation.

C.3 Ethiopia provisional coefficients (water HAZ, sanitation HAZ, post-12-month window)

Table C.2 — Ethiopia 2016 provisional HAZ coefficients under candidates C2 and C3 (post-12-month window, per 10 percentage points)

Candidate	Water β per 10	Sanitation β per 10
C2	−0.39	−0.45
C3	−0.39	−0.45

Note. Stored values from `docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv`, family `sens_ethiopia_PROVISIONAL`, outcome HAZ, window `post_12m`. Water (`W_IMP`): C2 `m0045`, −0.39 [−0.84, 0.06], p = .086, N = 6,568; C3 `m0046`, −0.39 [−0.84, 0.06], p = .088, N = 6,561. Sanitation (`S_IMP`): C2 `m0186`, −0.45 [−1.32, 0.41], p = .303, N = 6,568; C3 `m0187`, −0.45 [−1.31, 0.42], p = .312, N = 6,561. G = 586 clusters in all four models. Specification as for the established-date models (Section 4.6): cluster fixed effects, primary buffer with complete coverage, `PERWEIGHT` normalised within the estimation sample, reported age, wealth excluded. Standard errors are CRV1 clustered by DHS cluster; confidence intervals and p-values use a Student t distribution with G − 1 degrees of freedom. C1 estimates exist in the same file but are not shown, because C1 is invalid.

All four confidence intervals include zero. The estimates are reported here, and cross-referenced in Section 5.9, as an exploratory sensitivity with no bearing on the established-date pooled results in Section 5.6. Ethiopia is not pooled with Ghana, Kenya and Nigeria under any established-date label. The near-identical C2 and C3 coefficients follow from the near-identical samples in Section C.2 and do not validate either date. The gap between the two estimates is not a measure of how far either lies from an estimate based on the true interview dates.

C.4 Spatial-extraction and sensitivity diagnostics (established-date models, summarised)

The full 291-model register. The historical pipeline estimates 291 models in total, decomposing exactly into: 24 primary established-date cells (Section 5.6); 24 unweighted comparisons; 72 alternative-window models (Section 5.7: three further windows × 2 products × 3 outcomes × 4 sample groups); 24 wealth-adjustment, 24 containing-pixel, 24 rural-10km, 48 stricter-completion-rule (two postnatal windows), and 24 calendar-age-construction models (this section); 18 Ethiopia-provisional models (Sections C.1–C.3); and three 3-model overlap-exclusion families (9 total, Section 5.8). A separate 192-row file re-presents the window comparison on two sample bases (each window's own eligible sample and the sample common

to all four); its own-eligible-sample rows overlap with, rather than add to, the 291 already counted.

Each result row of Table C.3 changes one element of the main pooled specification and compares the pooled water–HAZ coefficient with the main estimate. The main estimate is model `m0004`: Ghana 2014, Kenya 2014 and Nigeria 2018 pooled, improved water (`W_IMP`), HAZ, post-12-month window, primary buffer with complete coverage, `PERWEIGHT` with equal country totals, reported age where delivered, wealth excluded, coefficient per 10 percentage points. Everything else is held fixed. Each sensitivity model, however, is estimated on its own eligible sample, so a difference from the main estimate can reflect a change in the children included as well as the change in specification.

Table C.3 — Spatial-extraction rule and sensitivity of the pooled established-date water–HAZ coefficient (per 10 percentage points)

Diagnostic	Result
Buffer coverage-completeness tolerance	Within 10^{-3} of full buffer area (valid-area fraction $\geq 1 - 10^{-3}$)
Spatial-method sensitivity (containing-pixel vs. buffer average)	Pooled water–HAZ: 0.68 vs. main 1.03
Rural 10 km buffer sensitivity (urban unchanged)	Pooled water–HAZ: 1.01 vs. main 1.03
Stricter postnatal completion rule (+1 month)	Pooled water–HAZ: 0.95 vs. main 1.03
Calendar-month age construction throughout	Pooled water–HAZ: 1.03 (materially unchanged)
Ghana excluded (DHS/IHME overlap; Kenya's water overlap remains)	Kenya and Nigeria pooled water–HAZ: 0.74 vs. three-country main 1.03

Note. Stored values from `docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv`, all HAZ, `W_IMP`, post-12-month window, weighted. Main `m0004` (primary): 1.03 [0.51, 1.54], $p < .001$, N = 23,018, G = 3,114. Containing pixel `m0028`: 0.68 [0.26, 1.10], $p = .002$, N = 23,642, G = 3,196. Rural 10 km `m0032`: 1.01 [0.42, 1.60], $p = .001$, N = 22,521, G = 3,062. Stricter completion `m0036`: 0.95 [0.43, 1.47], $p < .001$, N = 22,639, G = 3,113. Calendar age `m0044`: 1.03 [0.52, 1.53], $p < .001$, N = 23,018, G = 3,114. Ghana excluded `o003` (family `sens_overlap_excl_GH_water`, Kenya and Nigeria only): 0.74 [0.24, 1.24], $p = .004$, N = 21,078, G = 2,761. Inference as in Table C.2: CRV1 clustered by DHS cluster, $t(G - 1)$ confidence intervals and p-values.

Coverage rule and spatial assignment. The first row states a rule, not a result. For each cluster and year, the extraction computes the exact area of overlap between a disk of the stated radius and each raster pixel, after projecting the pixel's corners into a local azimuthal equidistant projection (Section 3.6). Pixels outside the raster, NoData pixels and values outside 0–100 are excluded from the valid area; they are never set to zero, and the buffer mean is taken over the valid area only. An annual value counts as complete when the valid area is at least $1 - 10^{-3}$ of the buffer's total area (`scripts/historical_wash/10_`

`spatial_extraction_v3.py`, `FULL_TOL`). A window is used only if every year it touches has a complete value within 2000–2017; nothing is extrapolated or filled. The primary buffer is 2 km for urban and 5 km for rural clusters. Partial-coverage values are kept only as a diagnostic status and enter none of the 291 stored models. The buffer is centred on the DHS-displaced cluster location, so it does not recover the true location. It is an area average of modelled coverage, not a population-weighted coverage estimate.

Containing pixel. This row replaces the buffer average with the value of the single pixel that contains the displaced cluster point. NoData gives a missing value, and no nearest-cell value is substituted. Because a valid value is needed only for that pixel, the eligible sample differs from the main one (N = 23,642 against 23,018). The move from 1.03 to 0.68 therefore reflects both the assignment rule and the sample.

Rural 10 km. Rural children are assigned the 10 km buffer average; urban children keep the 2 km primary buffer. The sample again differs (N = 22,521). A larger rural radius is a sensitivity to the size of the averaging area. It does not validate the DHS displacement probabilities or recover true locations.

Stricter completion. The postnatal completion rule is tightened from "interview month at least L months after the birth month" to at least L + 1 months, because the birth day is unknown. It applies only to the post-12-month and post-24-month windows; the table shows the post-12-month case. The estimation sample falls from 23,018 to 22,639.

Calendar age. The age control uses the calendar-month difference between interview and birth for all children instead of the reported completed age where it is delivered. In the established-date family, reported age is delivered only for Nigeria; Ghana and Kenya use calendar age in both versions. The sample is unchanged (N = 23,018), and the coefficient moves only in the third decimal (1.027 against 1.026).

Ghana excluded. Ghana's 2014 DHS round is an input to the IHME model for both products, and Kenya's 2014 round is an input for water (`docs/provenance/historical_wash_issue_register_v1_2026-10-07.md`, item 7). Removing Ghana leaves a two-country model pooled over Kenya and Nigeria (N = 21,078), not a three-country pooled model, and Kenya's water overlap remains in it. This row therefore shows the effect of removing Ghana's overlap, not an estimate free of DHS/IHME overlap. The stored file also contains a water model excluding both Ghana and Kenya; it leaves Nigeria alone and is a country-specific estimate, not a pooled one.

Weighting. Every row uses `PERWEIGHT` with equal country totals within its own estimation sample. Equal country totals do not give the countries equal influence on the pooled slope, which also depends on each country's residual exposure variation. The unweighted counterpart

of the main model (0.41) is reported in Section 5.6 and Appendix B. Neither the weighted nor the unweighted version corrects for selection into anthropometric measurement. The window and wealth sensitivities of the same family are shown in Section 5.7 and Section 5.8.

No sensitivity cell was selected or suppressed because of its significance. The sensitivity families were listed in the estimation specification (`docs/provenance/historical_wash_analysis_specification_v1_2026-10-06.md`, section 8). That record states that no coefficient had yet been estimated, and it was committed before the recorded time of the 291-model estimation run. The families are reported together in the methods note (`docs/provenance/historical_wash_preliminary_estimation_methods_2026-10-06.md`, section 7), and every one of the 291 models, including all outcomes and both products, is in the stored coefficient file. Table C.3 shows the pooled water–HAZ cells only.

What these diagnostics do not establish. All 291 historical models were estimated by `16_estimate_preliminary.py`. The 24 primary established-date cells, including the main model `m0004`, were independently reproduced with a second implementation, and two unweighted counterparts (`m0005`, `m0008`) were checked individually (Appendix B). The remaining 265 models, including every other row of Table C.3 and all 18 Ethiopia models, were not reproduced with a second implementation; they passed only internal checks. Agreement among these internally consistent diagnostics is not independent replication. Neither the reproduction nor the sensitivity results resolve causal identification, the undocumented weighting of the measured subsample, differential non-response among eligible children, the Nigeria 2018 measured-count discrepancy, the DHS/IHME input overlap, the use of current residence for early-life residence, coordinate displacement, the uncertainty of the modelled exposure surface, the uncertainty of Ethiopia's interview dates, or multiple testing, for which no adjustment has been made (Section 3.8, Section 5.6, Section 6). These remain qualifications of the research, not implementation errors.

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

Violations of this declaration may be considered an attempt at deception.

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Place, Date Signature